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(57) Abstract:

The present disclosure proposes a system (100) and method for processing structural brain MRI data for multi-stage Alzheimer's severity grading, clinical consistency validation, patient-specific severity-state transition estimation, and adaptive explainable neuroimaging. The system (100) comprises a computing device (102), a longitudinal patient data store 108, an MRI acquisition and preprocessing module (110), a deep neuroimaging feature encoder (112), a disease severity staging engine (114), a prediction uncertainty estimator (116), a clinical consistency validation engine (118), a patient-specific predictive NeuroTwin (120), a severity-state transition estimation engine (122), an adaptive explainability manager (124), a clinician decision interface (126), a communication interface (128), a network (130). The system (100) supports reproducible and auditable neuroimaging processing through model-version tracking, class-level performance documentation, structured processing records, and traceable longitudinal analysis information.

FORM 2

THE PATENT ACT, 1970

(39 of 1970)

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The Patent Rules, 2003

COMPLETE SPECIFICATION

(See section 10 and rule 13)

1. TITLE OF THE INVENTION:

System for MRI-Based Alzheimer's Severity Grading, Clinical Consistency Validation, Progression Forecasting, and Method Thereof

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3. PREAMBLE TO THE DESCRIPTION:

The following specification particularly describes the invention and the manner in which it is to be performed:

4. DESCRIPTION:

Field of the invention:

[0001] The present disclosure relates to the technical field of medical image processing and magnetic resonance imaging (MRI)-based neuroimaging analysis, and, more particularly, relates to a system and method for processing structural brain MRI data for multi-stage Alzheimer’s severity grading, clinical consistency validation, patient-specific severity-state transition estimation, and adaptive explainable neuroimaging.

Background of the invention:

[0002] Alzheimer’s disease represents a progressive neurodegenerative condition associated with structural and cognitive changes that require repeated clinical and neuroimaging assessment. Structural brain magnetic resonance imaging (MRI) provides non-invasive anatomical information for evaluation of cortical, hippocampal, ventricular, and related brain structures. Clinical assessment commonly relies on neuropsychological evaluation together with radiologist review of MRI acquisitions. Such workflows demand substantial specialist effort and remain susceptible to inter-rater variability. Automated MRI analysis therefore has technical relevance for standardized severity assessment, particularly across repeated examinations and subtle structural differences between adjacent severity states.

[0003] Conventional computational neuroimaging approaches apply convolutional neural networks, transformer-based models, transfer-learning frameworks, radiomics techniques, or multimodal feature-fusion architectures to brain-image analysis. Several approaches focus on binary or broad assessment categories, while four-class research models also exist. Difficulties remain in reliable differentiation of early or adjacent severity states, longitudinal comparison of repeated examinations, interpretation of model reliability, correlation between imaging-derived outputs and structured clinical information, and coordinated processing across clinical imaging infrastructure. Reviewed transfer-learning approaches further rely on model-specific or single-phase adaptation strategies, leaving robust adaptation of learned feature representations to structural brain MRI as a continuing technical challenge.

[0004] Indian Patent Publication IN202541130672A, titled "AI-driven intelligent agent system for automated MRI-based Alzheimer's disease detection and clinical triage," describes brain MRI acquisition and preprocessing followed by feature extraction and classification through a CNN-Transformer model. The reference further describes cognitively normal, mild cognitive impairment, and Alzheimer's disease categories, confidence-based risk assessment, Grad-CAM and saliency visualization, and clinical triage recommendations. The mapped disclosure remains directed principally toward present-state classification and review support and does not expressly address longitudinal cross-checking of successive MRI-derived severity outputs against structured patient-specific clinical parameters or coordinated linkage between historical severity assessments and subsequent state estimates.

[0005] Korean Patent Publication KR20240023327A, titled "Multimodal-based Alzheimer's progression prediction multitasking system and method," describes processing of clinical data and MRI data through feature-extraction paths followed by deep-learning-based feature fusion and progression-oriented analysis. Separate representations derived from clinical information and brain-image information undergo fusion for cognitive-impairment classification and Alzheimer's transition-risk assessment. The mapped architecture supports progression-oriented processing but does not expressly address longitudinal consistency across repeated severity assessments, quantified cross-source agreement between MRI-derived outputs and structured clinical parameters, or preservation of consistency and risk information across successive analysis cycles.

[0006] United States Patent Publication US20230057653A1, titled "Method and system and apparatus for quantifying uncertainty for medical image assessment," describes machine-learning-based medical-image assessment using imaging information, non-imaging information, information fusion, and modelling of uncertainty propagation. Such processing provides quantified uncertainty associated with automated medical assessment. The mapped disclosure addresses model-output uncertainty but does not expressly distinguish prediction uncertainty from disagreement between an imaging-derived severity output and structured patient-specific clinical information. A technical limitation therefore remains in assessment of cross-source consistency during repeated MRI-based severity evaluations.

[0007] The non-patent publication titled “Uncertainty-aware Alzheimer’s disease detection using Bayesian convolutional neural networks on MRI images” describes Bayesian convolutional neural networks for hierarchical MRI feature extraction together with epistemic and aleatoric uncertainty estimation. The approach generates class predictions with corresponding confidence information and further addresses cognitive-score prediction through multi-task learning. Uncertainty maps provide spatial information concerning regions associated with uncertain predictions. Such techniques improve reliability characterization at an individual model-output level but do not expressly address longitudinal comparison of successive severity outputs with prior imaging history and structured patient-specific clinical information.

[0008] The non-patent publication titled “BrainTwin-AI: a multimodal MRI-EEG-based cognitive digital twin for real-time brain health intelligence” describes fusion of MRI-derived structural representations with electroencephalography-derived functional information for generation of a brain-health risk representation. Gradient-weighted activation mapping provides spatial interpretation associated with model outputs. The reference demonstrates application of a digital-twin concept to neurological information. The mapped approach remains directed toward multimodal brain-health analysis rather than MRI-based multi-stage Alzheimer’s severity assessment coordinated with longitudinal clinical consistency evaluation across repeated examinations.

[0009] Taken together, the reviewed references demonstrate advances in MRI classification, multimodal progression-oriented analysis, uncertainty modelling, spatial attribution, and neurological digital-twin processing. Technical limitations remain in reliable separation of adjacent MRI-derived severity states, consistency of outputs across sequential examinations, cross-source correlation with structured clinical parameters, distinction between low model confidence and clinical discordance, generation of explanatory information under different reliability conditions, and preservation of temporally ordered patient-state information for later computational analysis.

[0010] Clinical neuroimaging deployments commonly involve MRI acquisition equipment, Picture Archiving and Communication Systems (PACS), computational analysis servers, hospital-edge resources, cloud-connected resources, clinician workstations, and Hospital

Information Systems (HIS). Task-specific approaches described in the reviewed materials do not provide a unified end-to-end workflow across these stages, which limits coordinated data flow across repeated assessment cycles. A technical requirement therefore remains for reliable interaction across modular clinical imaging environments while preserving standardized MRI-derived information, confidence information, clinical correlation data, and temporally ordered assessment history.

[0011] Accordingly, a need exists for an improved MRI-based neuroimaging processing system and associated method that address limitations related to variability in multi-stage severity assessment, isolated analysis of successive MRI examinations, inadequate correlation between imaging-derived outputs and structured clinical information, limited reliability characterization, fragmented longitudinal severity-state estimation, fixed explanatory processing, and disconnected clinical-system integration, while supporting technically coordinated operation across repeated assessment cycles.

[0012] By addressing all the above-mentioned problems, there is a need for a system and method for processing structural brain magnetic resonance imaging (MRI) data to generate multi-stage Alzheimer's severity probability information, cross-source clinical consistency information, patient-specific severity-state transition estimates, and confidence-responsive spatial attribution information. There is also a need for a system that standardizes structural brain MRI data through image conversion, intensity normalization, and spatial resampling for subsequent neuroimaging analysis. There is also a need for a system that extracts hierarchical neuroimaging feature representations through staged domain adaptation involving initial feature extraction and subsequent selective adaptation of a deep neuroimaging feature encoder to structural brain MRI data. There is also a need for a system that maps neuroimaging feature representations to multi-stage Alzheimer's severity probability information for characterization of MRI-derived severity states. There is also a need for a system that determines prediction confidence information from output-distribution characteristics, sampling variance, model disagreement, or one or more corresponding uncertainty indicators and generates an escalation indication upon satisfaction of a configurable confidence condition. There is also a need for a system that cross-evaluates MRI-derived severity information against structured patient-specific clinical parameters and generates quantified clinical consistency information together with a discordance indication based on a configurable consistency condition.

[0013] There is also a need for a system that creates and updates a Patient-Specific Predictive NeuroTwin storing an imaging timeline, severity-state history, predicted future states, clinical consistency history, treatment-response records, and severity-transition risk information across successive analysis cycles. There is also a need for a system that processes longitudinal severity-state history, assessment intervals, patient age, and structured clinical parameters to estimate future severity-state probabilities and corresponding transition-risk information. There is also a need for a system that selects a spatial attribution technique in response to prediction confidence information and generates MRI-based attribution evidence together with annotated anatomical-region information for interpretation of an MRI-derived severity output. There is also a need for a system that supports reduction of inter-rater variability and improvement of longitudinal consistency through coordinated processing of current MRI-derived severity information, prior assessment information, prediction confidence information, and structured patient-specific clinical parameters. There is also a need for a system that supports interoperable data exchange among MRI acquisition equipment, Picture Archiving and Communication Systems (PACS), neuroimaging analysis resources, clinician review interfaces, Hospital Information Systems (HIS), hospital-edge processing resources, and cloud-synchronization resources for longitudinal analysis and low-latency processing. There is also a need for a system that supports modular replacement of a neuroimaging feature encoder and integration of modality-specific feature streams derived from electroencephalography, speech, or eye-tracking data while maintaining interoperability with downstream severity-state, consistency, and longitudinal analysis operations. There is also a need for a system that supports reproducible and auditable neuroimaging processing through model-version tracking, class-level performance documentation, structured processing records, and traceable longitudinal analysis information.

Objectives of the invention:

[0014] The primary objective of the invention is to provide a system and method for processing structural brain magnetic resonance imaging (MRI) data to generate multi-stage

Alzheimer’s severity probability information, cross-source clinical consistency information, patient-specific severity-state transition estimates, and confidence-responsive spatial attribution information.

[0015] Another objective of the invention is to provide a system that standardizes structural brain MRI data through image conversion, intensity normalization, and spatial resampling for subsequent neuroimaging analysis.

[0016] The other objective of the invention is to provide a system that extracts hierarchical neuroimaging feature representations through staged domain adaptation involving initial feature extraction and subsequent selective adaptation of a deep neuroimaging feature encoder to structural brain MRI data.

[0017] The other objective of the invention is to provide a system that maps neuroimaging feature representations to multi-stage Alzheimer’s severity probability information for characterization of MRI-derived severity states.

[0018] The other objective of the invention is to provide a system that determines prediction confidence information from output-distribution characteristics, sampling variance, model disagreement, or combinations thereof and generates an escalation indication upon satisfaction of a configurable confidence condition.

[0019] The other objective of the invention is to provide a system that cross-evaluates MRI-derived severity information against structured patient-specific clinical parameters and generates quantified clinical consistency information together with a discordance indication based on a configurable consistency condition.

[0020] The other objective of the invention is to provide a system that creates and updates a Patient-Specific Predictive NeuroTwin storing an imaging timeline, severity-state history, predicted future states, clinical consistency history, treatment-response records, and severity-transition risk information across successive analysis cycles.

[0021] The other objective of the invention is to provide a system that processes longitudinal severity-state history, assessment intervals, patient age, and structured clinical parameters to estimate future severity-state probabilities and corresponding transition-risk information.

[0022] The other objective of the invention is to provide a system that selects a spatial attribution technique in response to prediction confidence information and generates MRI-based attribution evidence together with annotated anatomical-region information for interpretation of an MRI-derived severity output.

[0023] The other objective of the invention is to provide a system that supports reduction of inter-rater variability and improvement of longitudinal consistency by coordinating current MRI-derived severity information with prior assessment information, prediction confidence information, and structured patient-specific clinical parameters.

[0024] The other objective of the invention is to provide a system that supports interoperable data exchange among MRI acquisition equipment, Picture Archiving and Communication Systems (PACS), neuroimaging analysis resources, clinician review interfaces, Hospital Information Systems (HIS), hospital-edge processing resources, and cloud-synchronization resources for longitudinal analysis and low-latency processing.

[0025] Yet another objective of the invention is to provide a system that supports modular replacement of a neuroimaging feature encoder and integration of modality-specific feature streams derived from electroencephalography, speech, or eye-tracking data while maintaining interoperability with downstream severity-state, consistency, and longitudinal analysis operations.

[0026] Further, the objective of the invention is to provide a system that supports reproducible and auditable neuroimaging processing through model-version tracking, class-level performance documentation, structured processing records, and traceable longitudinal analysis information.

Summary of the invention:

[0027] The present disclosure proposes a system for MRI-based alzheimer’s severity grading, clinical consistency validation, progression forecasting, and method thereof. The following presents a simplified summary in order to provide a basic understanding of some aspects of the claimed subject matter. This summary is not an extensive overview. It is not intended to identify key/critical elements or to delineate the scope of the claimed subject matter. Its sole purpose is to present some concepts in a simplified form as a prelude to the more detailed description that is presented later.

[0028] In order to overcome the above deficiencies of the prior art, the present disclosure is to solve the technical problem to provide a system and method for processing structural brain MRI data for multi-stage Alzheimer's severity grading, clinical consistency validation, patient-specific severity-state transition estimation, and adaptive explainable neuroimaging.

[0029] According to one aspect, the invention provides a system for processing structural brain magnetic resonance imaging (MRI) data and maintaining patient-specific longitudinal neuroimaging state information. In one embodiment herein, the system comprises a computing device having at least one processor and a memory, a longitudinal patient data store, an MRI acquisition and preprocessing module, a deep neuroimaging feature encoder, a disease severity staging engine, a prediction uncertainty estimator, an adaptive explainability manager, a clinical consistency validation engine, a patient-specific predictive NeuroTwin, a severity-state transition estimation engine, and a clinician decision interface configured to perform MRI data processing, severity-state analysis, prediction confidence evaluation, clinical consistency validation, longitudinal patient-state management, future severity-state estimation, explainable neuroimaging generation, and structured decision-support generation operations.

[0030] In one embodiment herein, the MRI acquisition and preprocessing module receives structural brain magnetic resonance imaging (MRI) data from an imaging source and generates standardized neuroimaging input data through image conversion, intensity normalization, and spatial resampling operations. The MRI acquisition and preprocessing module processes Digital Imaging and Communications in Medicine (DICOM)-formatted structural brain MRI data through a Picture Archiving and Communication System (PACS) interface and provides standardized neuroimaging input data for subsequent feature extraction operations.

[0031] In one embodiment herein, the deep neuroimaging feature encoder processes the standardized neuroimaging input data and generates a hierarchical neuroimaging feature representation through a two-phase domain-adaptation protocol. The two-phase domain-adaptation protocol comprises a first training phase associated with frozen encoder layers and classification head training using feature representations generated by the frozen encoder layers, and a second training phase associated with selective adaptation of upper encoder layers to structural brain MRI data while maintaining frozen normalization layers. The deep neuroimaging feature encoder generates domain-adapted feature representations for severity-state analysis.

[0032] In one embodiment herein, the disease severity staging engine maps the hierarchical neuroimaging feature representation to a severity probability distribution across plurality of predetermined severity states and generates an MRI-derived severity-state output. The predetermined severity states correspond to Non-Demented, Very Mildly Demented, Mildly Demented, and Moderately Demented severity states. The disease severity staging engine generates structured severity-state information associated with processed structural brain MRI data.

[0033] In one embodiment herein, the prediction uncertainty estimator determines prediction confidence information from the severity probability distribution and compares the prediction confidence information with a confidence threshold. The prediction uncertainty estimator generates an escalation flag responsive to the prediction confidence information failing to satisfy the confidence threshold. The prediction uncertainty estimator derives prediction confidence information through uncertainty evaluation associated with entropy of the severity probability distribution, variance obtained through Monte Carlo sampling, disagreement among inference model outputs, or combinations thereof.

[0034] In one embodiment herein, the adaptive explainability manager receives prediction confidence information and dynamically selects a spatial attribution technique according to the prediction confidence information. The adaptive explainability manager generates spatial attribution information corresponding to the MRI-derived severity-state output. The spatial attribution technique comprises a gradient-weighted spatial attribution technique associated with a first confidence band, a score-based activation attribution technique associated with a second confidence band, and a layer-wise composite attribution technique associated with a third confidence band.

[0035] In one embodiment herein, the clinical consistency validation engine receives the MRI-derived severity-state output and structured patient-specific clinical parameters. The clinical consistency validation engine determines concordance between the MRI-derived severity-state output and the structured patient-specific clinical parameters and generates a clinical consistency score. The structured patient-specific clinical parameters comprise patient age, Mini-Mental State Examination (MMSE) score, Clinical Dementia Rating (CDR) score, prior MRI-derived severity state, clinical-history indicator, and apolipoprotein E (APOE) genotype indicator. The clinical consistency validation engine generates a discordance alert responsive to the clinical consistency score failing to satisfy a consistency threshold.

[0036] In one embodiment herein, the patient-specific predictive NeuroTwin maintains a structured longitudinal patient-state representation stored within the longitudinal patient data store. The patient-specific predictive NeuroTwin stores historical MRI analysis outcomes, longitudinal severity-state history, prediction confidence information, clinical-consistency records, treatment-response records, predicted future severity states, and severity-transition risk information. The patient-specific predictive NeuroTwin supports updating and retrieval of longitudinal patient-state information across successive MRI analysis cycles.

[0037] In one embodiment herein, the severity-state transition estimation engine processes a current MRI-derived severity-state output together with longitudinal information retrieved from the patient-specific predictive NeuroTwin. The severity-state transition estimation engine generates a future severity-state probability distribution and severity-transition risk information. The severity-state transition estimation engine stores a predicted future severity state within the patient-specific predictive NeuroTwin. The generated future severity-state information represents patient-specific severity-state transition information based on accumulated longitudinal information.

[0038] In one embodiment herein, the clinician decision interface consolidates the MRI-derived severity-state output, prediction confidence information, spatial attribution information, clinical consistency score, and predicted future severity state into a structured decision-support record for clinician review. The processor updates the patient-specific predictive NeuroTwin after an MRI analysis cycle using generated severity-state information, prediction confidence information, clinical consistency information, and predicted future severity-state information. The processor retrieves information from the updated patient-specific predictive NeuroTwin during a subsequent MRI analysis cycle

[0039] According to another aspect, the invention provides a computer-implemented method for training and operating the system for processing structural brain magnetic resonance imaging (MRI) data. In one embodiment herein, the method performs training of the deep neuroimaging feature encoder through the two-phase domain-adaptation protocol, receiving previously acquired structural brain MRI data through the MRI acquisition and preprocessing module, and generating standardized neuroimaging input data through image conversion, intensity normalization, and spatial resampling operations.

[0040] At one step, the deep neuroimaging feature encoder is trained through the two-phase domain-adaptation protocol by obtaining a deep convolutional, compound-scaled convolutional, attention-augmented convolutional, or attention-based feature encoder pretrained on a natural-image dataset. During a first training phase, encoder layers remain frozen and a classification head coupled to the feature encoder is trained using feature representations generated by the frozen encoder layers on a labelled structural brain MRI dataset. During a second training phase, upper encoder layers are selectively unfrozen and adapted to the structural brain MRI dataset while normalization layers remain frozen to generate a domain-adapted deep neuroimaging feature encoder.

[0041] At another step, previously acquired structural brain magnetic resonance imaging (MRI) data is received through the MRI acquisition and preprocessing module. At another step, the structural brain MRI data is processed through image conversion, intensity normalization, and spatial resampling operations to generate standardized neuroimaging input data. At another step, the standardized neuroimaging input data is processed through the domain-adapted deep neuroimaging feature encoder to generate a hierarchical neuroimaging feature representation.

[0042] At another step, the hierarchical neuroimaging feature representation is mapped through the disease severity staging engine to generate a severity probability distribution across plurality of predetermined severity states and an MRI-derived severity-state output.

At another step, prediction confidence information is determined from the severity probability distribution through the prediction uncertainty estimator. At another step, the prediction confidence information is compared with a confidence threshold and an escalation flag is generated responsive to the prediction confidence information failing to satisfy the confidence threshold.

[0043] At another step, a spatial attribution technique is dynamically selected through the adaptive explainability manager according to the prediction confidence information from plurality of distinct spatial attribution techniques, and spatial attribution information corresponding to the MRI-derived severity-state output is generated using the selected spatial attribution technique. At another step, the MRI-derived severity-state output and one or more structured patient-specific clinical parameters are received through the clinical consistency validation engine. At another step, concordance between the MRI-derived severity-state output and the structured patient-specific clinical parameters is determined.

[0044] At another step, a clinical consistency score representing the determined concordance is generated through the clinical consistency validation engine. At another step, the clinical consistency score is compared with a consistency threshold and a discordance alert is generated responsive to the clinical consistency score failing to satisfy the consistency threshold. At another step, longitudinal information associated with one or more preceding MRI analysis cycles is retrieved from the patient-specific predictive NeuroTwin stored in the longitudinal patient data store.

[0045] At another step, the MRI-derived severity-state output is processed together with the retrieved longitudinal information through the severity-state transition estimation engine to generate a future severity-state probability distribution and severity-transition risk information. At another step, a predicted future severity state is stored within the patient-specific predictive NeuroTwin. At another step, a structured neuroimaging analysis record is generated through the clinician decision interface by consolidating the MRI-derived severity-state output, the prediction confidence information, the spatial attribution information, the clinical consistency score, and the predicted future severity state. The structured neuroimaging analysis record is stored within the longitudinal patient data store and presented to a clinician.

[0046] At another step, the patient-specific predictive NeuroTwin is updated through at least one processor using the MRI-derived severity-state output, the prediction confidence information, the clinical consistency score, and the predicted future severity state. At another step, information is retrieved from the updated patient-specific predictive NeuroTwin during a subsequent MRI analysis cycle for processing structural brain MRI data associated with the subsequent MRI analysis cycle. Further, at another step, a structured escalation-protocol entry is generated responsive to the escalation flag or the discordance alert. The structured escalation-protocol entry identifies at least one of an additional-imaging action or a specialist-referral action.

[0047] In one embodiment, the spatial attribution technique is selected according to prediction confidence conditions. The gradient-weighted spatial attribution technique is selected responsive to prediction confidence information being at least 85 percent, a score-based activation attribution technique is selected responsive to prediction confidence information being from 70 percent to less than 85 percent, and a layer-wise composite attribution technique is selected responsive to prediction confidence information being below 70 percent.

[0048] In one embodiment, neuroimaging information and longitudinal patient information are exchanged through a communication interface and a network with one or more of the Picture Archiving and Communication System (PACS) interface, the Hospital Information System (HIS), the hospital-edge processing node, and the cloud-synchronization service.

[0049] Further, objects and advantages of the present invention will be apparent from a study of the following portion of the specification, the claims, and the attached drawings.

Detailed description of drawings:

[0050] The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate an embodiment of the invention, and, together with the description, explain the principles of the invention.

[0051] FIG. 1 illustrates a block diagram of a system for processing structural brain magnetic resonance imaging (MRI) data, generating severity-state information, maintaining longitudinal patient-state information, and providing clinical decision-support information, in accordance with an exemplary embodiment of the invention.

[0052] FIG. 2 illustrates a layered architecture of the system depicting an end-to-end neuroimaging processing workflow for generating MRI-derived severity-state outputs, prediction confidence information, clinical consistency information, longitudinal severity-state information, explainability information, and structured decision-support records, in accordance with an exemplary embodiment of the invention.

[0053] FIG. 3 illustrates a flow diagram of a neuroimaging analysis pipeline for processing structural brain MRI data, generating hierarchical neuroimaging feature representations, determining severity probability distribution, generating MRI-derived severity-state outputs, and producing explainable attribution information, in accordance with an exemplary embodiment of the invention.

[0054] FIG. 4 illustrates a flow diagram of an end-to-end MRI-based severity analysis workflow for generating standardized neuroimaging input data, severity-state information, prediction confidence information, clinical validation information, longitudinal patient-state information, and structured clinical decision-support information, in accordance with an exemplary embodiment of the invention.

[0055] FIG. 5 illustrates a bar chart representing validation dataset distribution across plurality of Alzheimer’s severity states, in accordance with an exemplary embodiment of the invention.

[0056] FIG. 6A illustrates representative structural brain MRI samples corresponding to a Non-Demented severity state, in accordance with an exemplary embodiment of the invention.

[0057] FIG. 6B illustrates representative structural brain MRI samples corresponding to a Very Mildly Demented severity state, in accordance with an exemplary embodiment of the invention.

[0058] FIG. 6C illustrates representative structural brain MRI samples corresponding to a Mildly Demented severity state, in accordance with an exemplary embodiment of the invention.

[0059] FIG. 6D illustrates representative structural brain MRI samples corresponding to a Moderately Demented severity state, in accordance with an exemplary embodiment of the invention.

[0060] FIG. 7 illustrates a clinical-system integration architecture for exchange of neuroimaging information, longitudinal patient information, and structured clinical decision-support records across interconnected healthcare computing environments, in accordance with an exemplary embodiment of the invention.

[0061] FIG. 8A illustrates a graphical representation of training and validation loss performance of a neuroimaging analysis system, in accordance with an exemplary embodiment of the invention.

[0062] FIG. 8B illustrates a graphical representation of training and validation accuracy performance of the neuroimaging analysis system, in accordance with an exemplary embodiment of the invention.

[0063] FIG. 8C illustrates a confusion matrix representing classification performance across plurality of severity states, in accordance with an exemplary embodiment of the invention.

[0064] FIG. 8D illustrates a graphical representation of receiver operating characteristic (ROC) performance corresponding to severity-state classification, in accordance with an exemplary embodiment of the invention.

[0065] FIG. 8E illustrates a graphical representation of precision-recall performance corresponding to severity-state classification, in accordance with an exemplary embodiment of the invention.

[0066] FIGs. 9A-9B illustrate a flowchart of a method for training and operating the system for processing structural brain MRI data, generating severity-state information, determining prediction confidence information, performing clinical consistency validation, generating future severity-state information, updating longitudinal patient-state information, and retrieving updated information during a subsequent MRI analysis cycle, in accordance with an exemplary embodiment of the invention.

Detailed invention disclosure:

[0067] Various embodiments of the present invention will be described in reference to the accompanying drawings. Wherever possible, same or similar reference numerals are used in the drawings and the description to refer to the same or like parts or steps.

[0068] The present disclosure has been made with a view towards solving the problem with the prior art described above, and it is an object of the present invention to provide a system and method for processing structural brain MRI data for multi-stage Alzheimer’s severity grading, clinical consistency validation, patient-specific severity-state transition estimation, and adaptive explainable neuroimaging.

[0069] According to one example embodiment of the invention, FIG. 1 refers to a block diagram of a system 100 for processing structural brain magnetic resonance imaging (MRI) data, generating severity-state information, maintaining longitudinal patient-state information, and providing clinical decision-support information. In one embodiment herein, the system 100 processes structural brain MRI data for multi-stage Alzheimer’s severity grading, clinical consistency validation, patient-specific severity-state transition estimation, and adaptive explainable neuroimaging. In one embodiment herein, the system 100 comprises a computing device 102, a longitudinal patient data store 108, an MRI acquisition and preprocessing module 110, a deep neuroimaging feature encoder 112, a disease severity staging engine 114, a prediction uncertainty estimator 116, a clinical consistency validation engine 118, a patient-specific predictive NeuroTwin 120, a severity-state transition estimation engine 122, an adaptive explainability manager 124, a clinician decision interface 126, a communication interface 128, a network 130, a Picture Archiving and Communication System (PACS) interface 132, a Hospital Information System (HIS) 134, a hospital-edge processing node 136, a cloud-synchronization service 138, and one or more modality-specific encoders 140.

[0070] In one embodiment herein, the computing device 102 provides a processing environment for executing operations associated with structural brain MRI analysis, severity-state generation, prediction confidence determination, clinical consistency validation, longitudinal patient-state management, explainable neuroimaging generation, and structured clinical decision-support generation. The computing device 102 comprises at least one processor 104 and a memory 106 operatively coupled to the at least one processor 104. The memory 106 stores executable instructions, processing parameters, model parameters, feature representations, MRI-derived severity-state information, prediction confidence information, clinical consistency information, longitudinal patient-state information, and structured decision-support records associated with operation of the system 100.

[0071] In one embodiment herein, the processor 104 refers to a hardware processing unit configured to execute operations associated with the system 100. The processor 104 comprises one or more processing units selected from a Central Processing Unit (CPU), Application-Specific Integrated Circuit (ASIC), Application-Specific Instruction-Set Processor (ASIP), Graphics Processing Unit (GPU), Physics Processing Unit (PPU), Digital Signal Processor (DSP), Field Programmable Gate Array (FPGA), Programmable Logic Device (PLD), controller, microcontroller unit, microprocessor, ARM-based processor, or combinations thereof. The processor 104 executes instructions stored within the memory 106 for controlling MRI data processing, neuroimaging feature extraction, severity-state analysis, uncertainty evaluation, clinical consistency validation, longitudinal patient-state updating, and decision-support generation operations.

[0072] In one embodiment herein, the memory 106 comprises one or more storage media configured for storing instructions and generated information associated with operation of the system 100. The memory 106 comprises one or more storage media selected from flash memory, hard disk storage, multimedia card memory, secure digital (SD) memory, extended data (XD) memory, random-access memory (RAM), static random-access memory (SRAM), read-only memory (ROM), electrically erasable programmable read-only memory (EEPROM), programmable read-only memory (PROM), magnetic storage memory, magnetic disk storage, optical disk storage, solid-state storage, or combinations thereof.

[0073] In one embodiment herein, the computing device 102 represents an electronic computing platform configured to execute neuroimaging analysis operations associated with the system 100. The computing device 102 comprises a server, workstation, desktop computer, laptop computer, tablet computer, portable computing device, healthcare computing terminal, edge computing device, Internet-of-Things (IoT) device, or combinations thereof. The computing device 102 supports local processing, edge-based processing, distributed processing, or combinations thereof according to deployment requirements of the healthcare computing environment.

[0074] In one embodiment herein, the system 100 utilizes artificial intelligence (AI)-based processing models for analysing structural brain MRI data and generating severity-state information. The AI-based processing models comprise machine learning (ML) models, deep learning (DL) models, neural network models, convolutional neural network (CNN) models, deep neural network (DNN) models, recurrent neural network (RNN) models, attention-based neural network models, transformer-based models, or combinations thereof. The deep neuroimaging feature encoder 112 utilizes an AI-based neural network architecture configured to extract hierarchical feature representations from standardized neuroimaging input data.

[0075] In one embodiment herein, the neural network architecture associated with the deep neuroimaging feature encoder 112 comprises plurality of computational layers configured to process standardized neuroimaging input data. The computational layers comprise convolutional layers, activation layers, average pooling layers, maximum pooling layers, concatenation layers, dropout layers, fully connected layers, SoftMax layers, attention layers, or combinations thereof. Each computational layer performs feature processing operations using associated weight values and coefficients for generating feature representations associated with structural brain MRI information.

[0076] In one embodiment herein, the neural network architecture associated with the deep neuroimaging feature encoder 112 comprises an input layer, one or more hidden layers, and an output layer. The input layer receives standardized neuroimaging input data and provides processed information to subsequent layers. The hidden layers transform received feature information into hierarchical representations through weighted computational operations and extract structural characteristics associated with the MRI data. The output layer provides generated feature representations utilized by the disease severity staging engine 114 for severity-state analysis.

[0077] In one embodiment herein, training of the deep neuroimaging feature encoder 112 is performed through one or more learning approaches selected from supervised learning, unsupervised learning, semi-supervised learning, reinforcement learning, regression-based learning, or combinations thereof. The deep neuroimaging feature encoder 112 is trained through the two-phase domain-adaptation protocol comprises a first training phase associated with frozen encoder layers and classification head optimization and a second training phase associated with selective adaptation of upper encoder layers while maintaining frozen normalization layers.

[0078] In one embodiment herein, the longitudinal patient data store 108 is accessible to the computing device 102 and stores longitudinal patient-specific information associated with successive MRI analysis cycles. The longitudinal patient data store 108 maintains historical MRI analysis outcomes, severity-state history, prediction confidence information, clinical-consistency records, treatment-response records, predicted future severity states, and severity-transition risk information. The longitudinal patient data store 108 supports retrieval of previously generated patient-state information during subsequent MRI analysis cycles.

[0079] In one embodiment herein, the MRI acquisition and preprocessing module 110 receives structural brain magnetic resonance imaging (MRI) data from an imaging source through clinical imaging infrastructure. The MRI acquisition and preprocessing module 110 processes received structural brain MRI data through image conversion, intensity normalization, and spatial resampling operations to generate standardized neuroimaging input data. In one exemplary implementation, the MRI acquisition and preprocessing module 110 receives Digital Imaging and Communications in Medicine (DICOM)-formatted MRI data through the Picture Archiving and Communication System (PACS) interface 132 and generates standardized neuroimaging input data for subsequent feature extraction operations.

[0080] In one embodiment herein, the MRI acquisition and preprocessing module 110 performs DICOM-to-tensor conversion operations for transforming received structural brain

MRI information into computationally processable neuroimaging input data. The MRI acquisition and preprocessing module 110 performs intensity normalization operations for reducing variation associated with MRI acquisition conditions and performs spatial resampling operations for generating standardized neuroimaging input data. The standardized neuroimaging input data is provided to the deep neuroimaging feature encoder 112 for generation of hierarchical neuroimaging feature representations.

[0081] In one embodiment herein, the deep neuroimaging feature encoder 112 processes the standardized neuroimaging input data and generates a hierarchical neuroimaging feature representation. The deep neuroimaging feature encoder 112 utilizes a two-phase domain-adaptation protocol for adapting learned feature representations to structural brain MRI data. The two-phase domain-adaptation protocol comprises a first training phase in which encoder layers remain frozen and a classification head coupled to the encoder layers is trained using feature representations generated by the frozen encoder layers, and a second training phase in which upper encoder layers are selectively unfrozen for adaptation to structural brain MRI data while normalization layers remain frozen. The deep neuroimaging feature encoder 112 generates domain-adapted hierarchical neuroimaging feature representations for subsequent severity-state analysis. In one exemplary implementation, the standardized neuroimaging input data comprises an input representation having dimensions of $224 \times 224 \times 3$, and the deep neuroimaging feature encoder 112 comprises an EfficientNetB0 backbone pretrained using an ImageNet dataset. Feature information generated by the deep neuroimaging feature encoder 112 undergoes global average pooling before being supplied to a classification head comprises a dense layer having 256 units and a rectified-linear-unit activation, a batch-normalization operation, a dropout operation having an exemplary dropout value of 0.4, and a SoftMax output corresponding to the plurality of predetermined severity states. The exemplary implementation provides one working architecture of the deep neuroimaging feature encoder 112 and disease severity staging engine 114, while the two-phase domain-adaptation protocol remains applicable to the alternative encoder architectures described herein.

[0082] In one embodiment herein, training of the deep neuroimaging feature encoder 112 is performed through a two-phase domain-adaptation regime. During the first training phase, the encoder layers remain fully frozen, the classification head is trained in a feature-extraction learning-rate regime, and early stopping is applied based on validation loss for stabilizing the classification head using feature representations generated by the frozen encoder layers. During the second training phase, upper encoder layers are selectively unfrozen and adapted to structural brain MRI data using a lower fine-tuning learning-rate regime while normalization layers remain frozen, and early stopping is applied based on validation loss. Keeping the normalization layers frozen during the second training phase maintains stable feature statistics during adaptation of the learned feature representations to the structural brain MRI domain. In one exemplary implementation, training-associated data augmentation is performed on structural brain MRI images through rotation within approximately plus or minus 20 degrees, horizontal reflection, zoom variation within approximately plus or minus 15 percent, width or height spatial shifting within approximately plus or minus 10 percent, and brightness variation within a range from 0.85 to 1.15. The augmentation operations are applied during training for improving generalization of the deep neuroimaging feature encoder 112 to variations present in structural brain MRI data.

[0083] In one embodiment herein, the disease severity staging engine 114 receives the hierarchical neuroimaging feature representation generated by the deep neuroimaging feature encoder 112 and generates an MRI-derived severity-state output. The disease severity staging engine 114 maps the hierarchical neuroimaging feature representation to a severity probability distribution associated with predetermined severity states. In one exemplary implementation, the severity probability distribution corresponds to Non-Demented, Very Mildly Demented, Mildly Demented, and Moderately Demented severity states.

[0084] In one embodiment herein, the prediction uncertainty estimator 116 determines prediction confidence information associated with the MRI-derived severity-state output. The prediction uncertainty estimator 116 evaluates the severity probability distribution and compares the prediction confidence information with a confidence threshold stored within the memory 106. The prediction uncertainty estimator 116 generates an escalation flag responsive to the prediction confidence information failing to satisfy the confidence threshold. The prediction confidence information is derived from one or more uncertainty indicators comprise entropy of the severity probability distribution, variance obtained through Monte Carlo sampling, disagreement among outputs generated by plurality of inference models, or combinations thereof.

[0085] In one embodiment herein, the clinical consistency validation engine 118 receives the MRI-derived severity-state output and one or more structured patient-specific clinical parameters. The clinical consistency validation engine 118 determines concordance between the MRI-derived severity-state output and the structured patient-specific clinical parameters and generates a clinical consistency score. The structured patient-specific clinical parameters comprise one or more of patient age, Mini-Mental State Examination (MMSE) score, Clinical Dementia Rating (CDR) score, prior MRI-derived severity state, clinical-history indicator, and apolipoprotein E (APOE) genotype indicator. The clinical consistency validation engine 118 generates a discordance alert responsive to the clinical consistency score failing to satisfy a consistency threshold. In one exemplary scoring implementation, the clinical consistency validation engine 118 generates, for each available structured patient-specific clinical parameter, a concordance value representing agreement between the respective structured patient-specific clinical parameter and an expected clinical profile corresponding to the MRI-derived severity-state output. The clinical consistency validation engine 118 applies a respective reliability weight to each generated concordance value and determines the clinical consistency score by weighted aggregation of the concordance values followed by normalization to a scoring range from 0 to 100. In one implementation, a concordance value comprises a binary concordance flag representing a concordant state or a discordant state. A clinical consistency score below a configurable consistency threshold, having an exemplary default value of 60, causes generation of the discordance alert.

[0086] In one embodiment herein, the clinical consistency validation engine 118 determines the clinical consistency score using the available structured patient-specific clinical parameters associated with the patient. For each available structured patient-specific clinical parameter, the clinical consistency validation engine 118 generates a binary concordance value representing a concordant state or a discordant state relative to an expected clinical profile corresponding to the MRI-derived severity-state output. A respective reliability weight associated with each available structured patient-specific clinical parameter is applied to the corresponding concordance value. The clinical consistency validation engine 118 determines a weighted aggregation of the concordance values and normalizes the weighted aggregation to a scoring range from 0 to 100 to obtain the clinical consistency score. In one exemplary implementation, a clinical consistency score below a configurable consistency threshold having a default value of 60 causes the clinical consistency validation engine 118 to generate the discordance alert. The clinical consistency validation engine 118 further generates a structured concordance report representing the clinical consistency score and the concordance evaluation associated with the available structured patient-specific clinical parameters for incorporation into the structured information presented through the clinician decision interface 126.

[0087] In one embodiment herein, the patient-specific predictive NeuroTwin 120 is maintained within the longitudinal patient data store 108 and provides a structured longitudinal patient-state representation. The patient-specific predictive NeuroTwin 120 stores historical MRI analysis outcomes, longitudinal severity-state history, prediction confidence information, clinical-consistency records, treatment-response records, predicted future severity states, and severity-transition risk information. The patient-specific predictive NeuroTwin 120 is updated after successive MRI analysis cycles for maintaining patient-specific longitudinal neuroimaging information. The patient-specific predictive NeuroTwin 120 represents the structured longitudinal patient-state representation maintained in the longitudinal patient data store 108, and is not limited by the nomenclature “NeuroTwin.” The patient-specific predictive NeuroTwin 120 comprises temporally associated patient-specific information including historical MRI analysis outcomes, longitudinal severity-state history, prediction confidence information, clinical-consistency records, treatment-response records, predicted future severity states, and severity-transition risk information, and is updated and subsequently retrieved during successive MRI analysis cycles.

[0088] In one embodiment herein, the severity-state transition estimation engine 122 processes a current MRI-derived severity-state output together with longitudinal information retrieved from the patient-specific predictive NeuroTwin 120. The severity-state transition estimation engine 122 processes current severity-state information, longitudinal severity-state history, assessment intervals, patient age, and structured patient-specific clinical parameters for generating a future severity-state probability distribution and severity-transition risk information. The generated future severity-state information is stored within the patient-specific predictive NeuroTwin 120 as a predicted future-state entry. In one exemplary implementation, the severity-state transition estimation engine 122 applies a probabilistic stage-transition model comprises a learned transition-probability matrix conditioned on patient-specific information comprises the current severity state, longitudinal severity-state history, time intervals between successive MRI analysis cycles, patient age, and one or more structured patient-specific clinical parameters. The severity-state transition estimation engine 122 determines probabilities corresponding to respective future severity states at a selected future time horizon and determines the severity-transition risk score as a probability of transition from the current severity state to a next or higher severity state within the selected future time horizon. In one exemplary implementation, the selected future time horizon is six months.

[0089] In one exemplary implementation, the severity-state transition estimation engine 122 retrieves a current severity state, a chronological longitudinal severity-state history, time intervals between successive MRI analysis cycles, patient age, and available structured patient-specific clinical parameters from the patient-specific predictive NeuroTwin 120. The severity-state transition estimation engine 122 applies a probabilistic stage-transition model comprises a learned transition-probability matrix conditioned on the retrieved patient-specific information to estimate respective probabilities corresponding to the predetermined severity states at a selected future time horizon. The severity-state transition estimation engine 122 determines a severity-transition risk score representing a probability of transition from the current severity state to a next or higher severity state within the selected future time horizon. In one exemplary implementation, the selected future time horizon is six months. The generated future-state information comprises probabilities corresponding to the Non-Demented, Very Mildly Demented, Mildly Demented, and Moderately Demented severity states together with the severity-transition risk score. The future-state information is stored as a predicted future-state entry in the patient-specific predictive NeuroTwin 120 and is usable for generating longitudinal severity-transition information and a structured risk representation for presentation through the clinician decision interface 126.

[0090] In one embodiment herein, the adaptive explainability manager 124 receives prediction confidence information from the prediction uncertainty estimator 116 and selects a spatial attribution technique according to the prediction confidence information. The adaptive explainability manager 124 generates spatial attribution information corresponding to the MRI-derived severity-state output. The selected spatial attribution technique comprises a gradient-weighted spatial attribution technique, a score-based activation attribution technique, or a layer-wise composite attribution technique according to corresponding prediction confidence bands.

[0091] In one exemplary implementation, the adaptive explainability manager 124 selects the spatial attribution technique according to the prediction confidence information such that a gradient-weighted spatial attribution technique is selected when the prediction confidence information is at least 85 percent, a score-based activation attribution technique is selected when the prediction confidence information ranges from 70 percent to less than 85 percent, and a layer-wise composite attribution technique is selected when the prediction confidence information is below 70 percent. Selection of the spatial attribution technique is therefore responsive to a reliability condition associated with the MRI-derived severity-state output. The spatial attribution information generated by the adaptive explainability manager 124 comprises a clinical evidence representation having the structural brain MRI image, a spatial attribution map corresponding to the MRI-derived severity-state output, and an annotated anatomical-region overlay. In one exemplary implementation, anatomical-region information is associated with high-attribution spatial locations for identifying one or more corresponding brain regions that comprise the hippocampus, temporal lobe, periventricular regions, or cortical surface. The generated spatial attribution information is provided together with the corresponding prediction confidence information for clinician review through the clinician decision interface 126.

[0092] In one embodiment herein, the clinician decision interface 126 consolidates MRI-derived severity-state output information, prediction confidence information, spatial attribution information, clinical consistency information, and predicted future severity-state information into a structured decision-support record. The clinician decision interface 126 generates structured neuroimaging analysis information for clinician review and supports integration with healthcare information environments.

[0093] In one embodiment herein, the structured decision-support record generated through the clinician decision interface 126 is provided as a structured evidence package for clinician review. The structured decision-support record consolidates the MRI-derived severity-state output, severity probability information, prediction confidence information, spatial attribution information, clinical consistency information, future severity-state information, and longitudinal patient-state information for review by a radiologist or other clinician. The system 100 provides the generated information as clinical decision-support information and the at least one processor 104 does not generate an autonomous treatment recommendation based on the generated information. A discordance alert or escalation flag generated through the system 100 is incorporated into the structured decision-support record for clinician review. Where an escalation-protocol entry is generated, the escalation-protocol entry identifies an additional-imaging action or a specialist-referral action as structured information within the decision-support record, and the corresponding record is presented through the clinician decision interface 126 for clinical review.

[0094] In one embodiment herein, the communication interface 128 exchanges neuroimaging information and longitudinal patient information through the network 130. The network 130 supports wired communication networks, wireless communication networks, or combinations thereof. Examples of wired communication networks comprise Local Area Network (LAN), Wide Area Network (WAN), Ethernet-based networks, and healthcare communication networks. Examples of wireless communication networks comprise cellular networks, wireless LAN (Wi-Fi), Bluetooth, Bluetooth Low Energy (BLE), Zigbee, Wi-Fi Direct (WFD), Ultra-wideband (UWB), infrared communication, and near field communication (NFC).

[0095] In one embodiment herein, the communication interface 128 supports exchange of neuroimaging information, longitudinal patient information, and structured decision-support records through communication protocols associated with the network 130. The communication interface 128 supports communication through Bluetooth Low Energy (BLE), Wi-Fi, Zigbee, Wi-Fi Direct (WFD), Ultra-wideband (UWB), infrared data association (IrDA), near field communication (NFC), cellular communication, or combinations thereof.

[0096] In one embodiment herein, the communication interface 128 supports communication between the system 100 and healthcare computing environments associated with the PACS interface 132, Hospital Information System (HIS) 134, hospital-edge processing node 136, and cloud-synchronization service 138. The communication interface 128 facilitates transfer and synchronization of structural brain MRI information, MRI-derived severity-state outputs, prediction confidence information, clinical consistency information, severity-transition information, spatial attribution information, and structured decision-support records.

[0097] In one embodiment herein, the hospital-edge processing node 136 provides a local processing environment for execution of neuroimaging analysis operations associated with the system 100. The hospital-edge processing node 136 supports processing of structural brain MRI information within a healthcare computing environment and facilitates generation of structured neuroimaging analysis information for presentation through the clinician decision interface 126.

[0098] In one embodiment herein, the cloud-synchronization service 138 maintains synchronization of longitudinal patient-state information associated with the patient-specific predictive NeuroTwin 120. The cloud-synchronization service 138 supports storage and retrieval of longitudinal patient information comprises historical MRI analysis outcomes, severity-state history, prediction confidence information, clinical-consistency records, predicted future severity states, and severity-transition risk information.

[0099] In one embodiment herein, the one or more modality-specific encoders 140 provide additional feature information derived from electroencephalography (EEG) data, speech-derived data, or eye-tracking data. The modality-specific encoders 140 provide additional feature information to the patient-specific predictive NeuroTwin 120 while maintaining interoperability with the disease severity staging engine 114, the clinical consistency validation engine 118, and the severity-state transition estimation engine 122.

[0100] In one embodiment herein, the coordinated operation of the MRI acquisition and preprocessing module 110, the deep neuroimaging feature encoder 112, the disease severity staging engine 114, the prediction uncertainty estimator 116, the clinical consistency validation engine 118, the patient-specific predictive NeuroTwin 120, the severity-state transition estimation engine 122, the adaptive explainability manager 124, and the clinician decision interface 126 establishes an integrated neuroimaging analysis framework. The system 100 generates MRI-derived severity-state information, confidence-responsive explainability information, clinical consistency information, longitudinal severity-state information, and structured decision-support information during successive MRI analysis cycles.

[0101] According to another example embodiment of the invention, FIG. 2 refers to a layered architecture 200 of the system 100 depicting an end-to-end neuroimaging processing workflow for generating MRI-derived severity-state outputs, prediction confidence information, clinical consistency information, longitudinal severity-state information, explainability information, and structured decision-support records. In one embodiment herein, the layered architecture 200 represents a modular technical architecture integrating neuroimaging data processing, prediction confidence evaluation, clinical consistency validation, severity-state transition estimation, adaptive explainability generation, longitudinal patient-state management, and structured decision-support generation within the system 100. In one embodiment herein, the layered architecture 200 comprises an MRI acquisition and preprocessing layer associated with the MRI acquisition and preprocessing module 110, a deep neuroimaging feature encoding layer associated with the deep neuroimaging feature encoder 112, a disease severity staging layer associated with the disease severity staging engine 114, a prediction uncertainty estimation layer associated with the prediction uncertainty estimator 116, an adaptive explainability layer associated with the adaptive explainability manager 124, a clinical consistency validation layer associated with the clinical consistency validation engine 118, a severity-state transition estimation layer associated with the severity-state transition estimation engine 122, a patient-specific predictive NeuroTwin layer associated with the patient-specific predictive NeuroTwin 120, and a clinician decision-support layer associated with the clinician decision interface 126.

[0102] The modules of the layered architecture 200 are independently operable and replaceable while maintaining interoperability through structured exchange of generated information. The modules communicate through a structured data protocol configured to exchange MRI-derived severity-state information, prediction confidence information, clinical consistency information, longitudinal patient-state information, explainability information, and structured decision-support information. In one embodiment herein, the structured data protocol of the layered architecture 200 provides machine-readable exchange of standardized neuroimaging input data, hierarchical neuroimaging feature representations, MRI-derived severity-state information, prediction confidence information, clinical consistency information, longitudinal patient-state information, future severity-state information, spatial attribution information, and structured decision-support information between corresponding modules of the system 100. The structured exchange enables generated information from a current MRI analysis cycle to be stored in the longitudinal patient data store 108 and retrieved during a subsequent MRI analysis cycle.

[0103] In one implementation, DICOM-formatted structural brain MRI information is received through the PACS interface 132 and transferred for neuroimaging processing through the computing device 102 or the hospital-edge processing node 136. Generated longitudinal patient-state information is synchronizable through the cloud-synchronization service 138, and structured severity-state and clinical evidence information is transferable through the clinician decision interface 126 for storage within the Hospital Information System 134. The coordinated architecture thereby provides an end-to-end machine-processing path between clinical imaging acquisition, standardized MRI preprocessing, neuroimaging feature generation, severity-state processing, longitudinal patient-state updating, structured clinician review, and hospital-information storage.

[0104] The MRI acquisition and preprocessing module 110 receives structural brain magnetic resonance imaging (MRI) data from an imaging source and generates standardized neuroimaging input data. The MRI acquisition and preprocessing module 110 performs image conversion, intensity normalization, spatial resampling, and training-associated data augmentation operations for preparing structural brain MRI data for subsequent neuroimaging processing. In one exemplary implementation, the MRI acquisition and preprocessing module 110 receives Digital Imaging and Communications in Medicine (DICOM)-formatted MRI data through the Picture Archiving and Communication System (PACS) interface 132.

[0105] The deep neuroimaging feature encoder 112 processes the standardized neuroimaging input data and generates hierarchical neuroimaging feature representations. The deep neuroimaging feature encoder 112 is trained through a two-phase domain-adaptation protocol for adapting feature representations to structural brain MRI data. The two-phase domain-adaptation protocol comprises a first training phase associated with frozen encoder layers and classification head training using feature representations generated by the frozen encoder layers, and a second training phase associated with selective adaptation of upper encoder layers to structural brain MRI data while maintaining frozen normalization layers. The deep neuroimaging feature encoder 112 generates domain-adapted hierarchical neuroimaging feature representations for subsequent severity-state processing.

[0106] The disease severity staging engine 114 receives the hierarchical neuroimaging feature representation generated by the deep neuroimaging feature encoder 112 and generates an MRI-derived severity-state output. The disease severity staging engine 114 maps the hierarchical neuroimaging feature representation to a severity probability distribution associated with plurality of predetermined severity states. In one exemplary implementation, the disease severity staging engine 114 generates probability information corresponding to Non-Demented, Very Mildly Demented, Mildly Demented, and Moderately Demented severity states.

[0107] The prediction uncertainty estimator 116 determines prediction confidence information associated with the MRI-derived severity-state output. The prediction uncertainty estimator 116 evaluates the severity probability distribution and generates an escalation flag based on comparison of the prediction confidence information with a confidence threshold. The prediction confidence information is derived from one or more uncertainty indicators comprise entropy of the severity probability distribution, variance obtained through Monte Carlo sampling, disagreement among inference model outputs, or combinations thereof.

[0108] The adaptive explainability manager 124 receives the prediction confidence information generated by the prediction uncertainty estimator 116 and generates spatial attribution information corresponding to the MRI-derived severity-state output. The adaptive explainability manager 124 selects a spatial attribution technique according to the prediction confidence information. The selected spatial attribution technique comprises a gradient-weighted spatial attribution technique, a score-based activation attribution technique, or a layer-wise composite attribution technique according to corresponding prediction confidence bands.

[0109] The clinical consistency validation engine 118 receives the MRI-derived severity-state output and structured patient-specific clinical parameters. The clinical consistency validation engine 118 determines concordance between the MRI-derived severity-state output and the structured patient-specific clinical parameters and generates a clinical consistency score. The structured patient-specific clinical parameters comprise one or more of patient age, Mini-Mental State Examination (MMSE) score, Clinical Dementia Rating (CDR) score, prior MRI-derived severity state, clinical-history indicator, and apolipoprotein E (APOE) genotype indicator. The clinical consistency validation engine 118 generates a discordance alert responsive to the clinical consistency score failing to satisfy a consistency threshold.

[0110] The severity-state transition estimation engine 122 processes the MRI-derived severity-state output together with longitudinal information retrieved from the patient-specific predictive NeuroTwin 120. The severity-state transition estimation engine 122 processes current severity-state information, longitudinal severity-state history, assessment intervals, patient age, and structured clinical parameters for generating a future severity-state probability distribution and severity-transition risk information. The generated future severity-state information is stored within the patient-specific predictive NeuroTwin 120 as a predicted future-state entry.

[0111] The patient-specific predictive NeuroTwin 120 maintains a structured longitudinal patient-state representation associated with successive MRI analysis cycles. The patient-specific predictive NeuroTwin 120 stores historical MRI analysis outcomes, longitudinal severity-state history, prediction confidence information, clinical-consistency records, treatment-response records, predicted future severity states, and severity-transition risk information. The patient-specific predictive NeuroTwin 120 supports updating and retrieval of longitudinal patient-state information during subsequent MRI analysis cycles.

[0112] The clinician decision interface 126 consolidates MRI-derived severity-state output information, prediction confidence information, spatial attribution information, clinical consistency information, and predicted future severity-state information into a structured decision-support record. The clinician decision interface 126 generates structured neuroimaging analysis information for clinician review and supports integration with healthcare information environments.

[0113] The communication interface 128 facilitates exchange of neuroimaging information and longitudinal patient information through the network 130. The communication interface 128 supports communication with one or more of the PACS interface 132, Hospital Information System (HIS) 134, hospital-edge processing node 136, and cloud-synchronization service 138 for exchange and management of structured neuroimaging analysis information and longitudinal patient-state information.

[0114] The one or more modality-specific encoders 140 provide additional feature information derived from electroencephalography (EEG) data, speech-derived data, or eye-tracking data. The modality-specific encoders 140 provide additional feature information to the patient-specific predictive NeuroTwin 120 while maintaining interoperability with the disease severity staging engine 114, the clinical consistency validation engine 118, and the severity-state transition estimation engine 122.

[0115] The layered architecture 200 establishes a modular neuroimaging processing framework integrating structural brain MRI preprocessing, domain-adapted feature extraction, severity-state analysis, prediction confidence evaluation, clinical consistency assessment, longitudinal patient-state maintenance, adaptive explainability generation, and structured clinical decision-support generation within the system 100. The layered architecture 200 supports successive MRI analysis cycles through maintenance, updating, and retrieval of patient-specific longitudinal information.

[0116] According to another example embodiment of the invention, FIG. 3 refers to a flow diagram 300 of a neuroimaging analysis pipeline for processing structural brain MRI data, generating hierarchical neuroimaging feature representations, determining severity probability distribution, generating MRI-derived severity-state outputs, and producing explainable attribution information. In one embodiment herein, the flow diagram 300 represents an operational sequence of the system 100 for transforming structural brain MRI data into severity-state information and associated explainability information through coordinated operation of the MRI acquisition and preprocessing module 110, the deep neuroimaging feature encoder 112, the disease severity staging engine 114, the prediction uncertainty estimator 116, and the adaptive explainability manager 124.

[0117] At step 302, the deep neuroimaging feature encoder 112 is trained through a two-phase domain-adaptation protocol for adapting feature representations to structural brain MRI data. In one embodiment herein, the deep neuroimaging feature encoder 112 is obtained from a deep convolutional, compound-scaled convolutional, attention-augmented convolutional, or attention-based feature encoder pretrained on a natural-image dataset. During a first training phase, encoder layers of the deep neuroimaging feature encoder 112 remain frozen and a classification head coupled to the encoder layers is trained using feature representations generated by the frozen encoder layers. During a second training phase, upper encoder layers of the deep neuroimaging feature encoder 112 are selectively unfrozen for adaptation to structural brain MRI data while normalization layers remain frozen, thereby generating a domain-adapted deep neuroimaging feature encoder 112.

[0118] At step 304, previously acquired structural brain magnetic resonance imaging (MRI) data is received through the MRI acquisition and preprocessing module 110. In one embodiment herein, the MRI acquisition and preprocessing module 110 receives structural brain MRI data from an imaging source through clinical imaging infrastructure for subsequent neuroimaging processing.

[0119] At step 306, the MRI acquisition and preprocessing module 110 performs image conversion, intensity normalization, and spatial resampling on the structural brain MRI data to generate standardized neuroimaging input data. In one exemplary implementation, the MRI acquisition and preprocessing module 110 processes Digital Imaging and Communications in Medicine (DICOM)-formatted MRI data received through a Picture Archiving and Communication System (PACS) interface 132 and generates standardized neuroimaging input data for feature extraction.

[0120] At step 308, the domain-adapted deep neuroimaging feature encoder 112 processes the standardized neuroimaging input data and generates a hierarchical neuroimaging feature representation. The hierarchical neuroimaging feature representation represents extracted structural characteristics associated with the processed MRI data for subsequent severity-state processing.

[0121] At step 310, the disease severity staging engine 114 processes the hierarchical neuroimaging feature representation and generates an MRI-derived severity-state output. In one embodiment herein, the disease severity staging engine 114 maps the hierarchical neuroimaging feature representation to a severity probability distribution associated with plurality of predetermined severity states. In one exemplary implementation, the severity probability distribution corresponds to Non-Demented, Very Mildly Demented, Mildly Demented, and Moderately Demented severity states.

[0122] At step 312, the prediction uncertainty estimator 116 determines prediction confidence information associated with the MRI-derived severity-state output. In one embodiment herein, the prediction uncertainty estimator 116 evaluates the severity probability distribution through one or more uncertainty indicators comprise entropy of the severity probability distribution, variance obtained through Monte Carlo sampling, disagreement among inference model outputs, or combinations thereof. The prediction uncertainty estimator 116 generates an escalation flag responsive to the prediction confidence information failing to satisfy a confidence threshold.

[0123] At step 314, the adaptive explainability manager 124 selects a spatial attribution technique according to the prediction confidence information and generates spatial attribution information corresponding to the MRI-derived severity-state output. In one embodiment herein, the adaptive explainability manager 124 selects a gradient-weighted spatial attribution technique, a score-based activation attribution technique, or a layer-wise composite attribution technique according to corresponding prediction confidence bands. The generated spatial attribution information provides spatially associated attribution information corresponding to the MRI-derived severity-state output.

[0124] The flow diagram 300 establishes a neuroimaging analysis pipeline for generating MRI-derived severity-state outputs through domain-adapted feature extraction, severity probability generation, prediction confidence evaluation, and confidence-responsive explainability generation. The generated information is utilized by subsequent modules of the system 100 for clinical consistency validation, longitudinal patient-state management, severity-state transition estimation, and structured decision-support generation.

[0125] According to another example embodiment of the invention, FIG. 4 refers to a flow diagram 400 of an end-to-end MRI-based severity analysis workflow for generating standardized neuroimaging input data, severity-state information, prediction confidence information, clinical consistency information, longitudinal patient-state information, and structured clinical decision-support information. In one embodiment herein, the flow diagram 400 represents an operational workflow of the system 100 for processing structural brain magnetic resonance imaging (MRI) data, validating MRI-derived severity-state information, maintaining patient-specific longitudinal information, generating future severity-state information, and producing structured decision-support information through coordinated operation of the MRI acquisition and preprocessing module 110, the deep neuroimaging feature encoder 112, the disease severity staging engine 114, the prediction uncertainty estimator 116, the clinical consistency validation engine 118, the patient-specific predictive NeuroTwin 120, the severity-state transition estimation engine 122, the adaptive explainability manager 124, and the clinician decision interface 126.

[0126] At step 402, the system 100 receives previously acquired structural brain magnetic resonance imaging (MRI) data through the MRI acquisition and preprocessing module 110. In one embodiment herein, the MRI acquisition and preprocessing module 110 receives structural brain MRI data from an imaging source and generates DICOM-formatted neuroimaging input information for subsequent processing.

[0127] At step 404, the MRI acquisition and preprocessing module 110 performs preprocessing operations on the structural brain MRI data to generate standardized neuroimaging input data. The preprocessing operations comprise DICOM-to-tensor conversion, intensity normalization, and spatial resampling for preparing the structural brain MRI data for subsequent neuroimaging feature extraction.

[0128] At step 406, the deep neuroimaging feature encoder 112 processes the standardized neuroimaging input data and generates a hierarchical neuroimaging feature representation. In one embodiment herein, the deep neuroimaging feature encoder 112 operates as a domain-adapted feature extraction module trained through a two-phase domain-adaptation protocol. The two-phase domain-adaptation protocol comprises a first training phase associated with frozen encoder layers and classification head training using feature representations generated by the frozen encoder layers, and a second training phase associated with selective adaptation of upper encoder layers to structural brain MRI data while normalization layers remain frozen.

[0129] At step 408, the disease severity staging engine 114 processes the hierarchical neuroimaging feature representation and generates an MRI-derived severity-state output. The disease severity staging engine 114 generates a severity probability distribution associated with plurality of predetermined severity states. In one exemplary implementation, the severity probability distribution corresponds to Non-Demented, Very Mildly Demented, Mildly Demented, and Moderately Demented severity states.

[0130] At step 410, the prediction uncertainty estimator 116 determines prediction confidence information associated with the MRI-derived severity-state output and evaluates the prediction confidence information against a confidence threshold. The prediction uncertainty estimator 116 generates an escalation flag responsive to the prediction confidence information failing to satisfy the confidence threshold. The adaptive explainability manager 124 selects a spatial attribution technique according to the prediction confidence information and generates spatial attribution information corresponding to the MRI-derived severity-state output. The selected spatial attribution technique comprises a gradient-weighted spatial attribution technique, a score-based activation attribution technique, or a layer-wise composite attribution technique according to corresponding prediction confidence conditions.

[0131] At step 412, the clinical consistency validation engine 118 receives the MRI-derived severity-state output and structured patient-specific clinical parameters and determines concordance between the MRI-derived severity-state output and the structured patient-specific clinical parameters. The clinical consistency validation engine 118 generates a clinical consistency score and a discordance alert based on comparison with a consistency threshold. The severity-state transition estimation engine 122 processes longitudinal information retrieved from the patient-specific predictive NeuroTwin 120 together with the MRI-derived severity-state output and generates a future severity-state probability distribution and severity-transition risk information. The patient-specific predictive NeuroTwin 120 stores imaging timeline information, longitudinal severity-state history, prediction confidence information, clinical consistency information, treatment-response records, predicted future severity states, and severity-transition risk information. The clinician decision interface 126 consolidates the MRI-derived severity-state output, prediction confidence information, spatial attribution information, clinical consistency information, and predicted future severity-state information into a structured decision-support record.

[0132] The flow diagram 400 establishes an end-to-end MRI-based severity analysis workflow integrating MRI data acquisition, preprocessing, domain-adapted neuroimaging feature extraction, severity-state generation, prediction confidence evaluation, confidence-responsive explainability generation, clinical consistency validation, longitudinal NeuroTwin updating, future severity-state estimation, and structured clinical decision-support generation within the system 100. The workflow supports repeated MRI analysis cycles through maintenance, updating, and retrieval of patient-specific longitudinal information.

[0133] According to another example embodiment of the invention, FIG. 5 refers to a bar chart 500 representing validation dataset distribution across plurality of Alzheimer’s severity states. In one embodiment herein, the bar chart 500 represents distribution of structural brain magnetic resonance imaging (MRI) images utilized for training, validation, and test performance evaluation of the system 100. The bar chart 500 represents distribution of MRI images across Non-Demented, Very Mildly Demented, Mildly Demented, and Moderately Demented severity states utilized for evaluation of MRI-derived severity-state classification performance generated through the disease severity staging engine 114.

[0134] In one embodiment herein, the bar chart 500 represents a dataset distribution comprises approximately 12,804 MRI images associated with the Non-Demented severity state corresponding to approximately 29.1% of the dataset, approximately 11,176 MRI images associated with the Very Mildly Demented severity state corresponding to approximately 25.4% of the dataset, approximately 10,000 MRI images associated with the Mildly Demented severity state corresponding to approximately 22.7% of the dataset, and approximately 10,000 MRI images associated with the Moderately Demented severity state corresponding to approximately 22.7% of the dataset. The represented dataset distribution provides evaluation information associated with classification performance across the plurality of predetermined severity states processed through the system 100.

[0135] In one embodiment herein, the structural brain MRI dataset represented through the bar chart 500 is utilized for evaluating operation of the deep neuroimaging feature encoder 112 and the disease severity staging engine 114. The distribution information supports assessment of hierarchical neuroimaging feature generation, severity probability distribution generation, and MRI-derived severity-state output generation across Non-Demented, Very Mildly Demented, Mildly Demented, and Moderately Demented severity states. The bar chart 500 provides validation dataset distribution information associated with performance evaluation of the neuroimaging analysis system 100.

[0136] According to another example embodiment of the invention, FIG. 6A refers to representative structural brain MRI samples 600 corresponding to a Non-Demented severity state. In one embodiment herein, the representative structural brain MRI samples 600 represent structural brain MRI characteristics associated with the Non-Demented severity state utilized for evaluation of MRI-derived severity-state classification performed through the system 100. The representative structural brain MRI samples 600 represent brain MRI information associated with normal brain morphology without visible cortical atrophy or ventricle enlargement. The representative structural brain MRI samples 600 provide reference information associated with structural brain MRI data received through the MRI acquisition and preprocessing module 110 and processed through the deep neuroimaging feature encoder 112 and the disease severity staging engine 114 for generating severity-state outputs.

[0137] According to another example embodiment of the invention, FIG. 6B refers to representative structural brain MRI samples 602 corresponding to a Very Mildly Demented severity state. In one embodiment herein, the representative structural brain MRI samples 602 represent structural brain MRI characteristics associated with the Very Mildly Demented severity state utilized for evaluation of early-stage severity-state classification performed through the system 100. The representative structural brain MRI samples 602 represent structural brain MRI information associated with subtle early-stage structural changes and minimal sulcal widening. The representative structural brain MRI samples 602 provide reference information associated with early-stage MRI feature processing performed through the MRI acquisition and preprocessing module 110, hierarchical feature generation through the deep neuroimaging feature encoder 112, and severity-state classification through the disease severity staging engine 114.

[0138] According to another example embodiment of the invention, FIG. 6C refers to representative structural brain MRI samples 604 corresponding to a Mildly Demented severity state. In one embodiment herein, the representative structural brain MRI samples 604 represent structural brain MRI characteristics associated with the Mildly Demented severity state utilized for evaluation of intermediate severity-state classification performed through the system 100. The representative structural brain MRI samples 604 represent structural brain MRI information associated with moderate cortical atrophy, widened sulci, and early hippocampal volume loss. The representative structural brain MRI samples 604 provide reference information associated with generation of hierarchical neuroimaging feature representations through the deep neuroimaging feature encoder 112 and generation of MRI-derived severity-state outputs through the disease severity staging engine 114.

[0139] According to another example embodiment of the invention, FIG. 6D refers to representative structural brain MRI samples 606 corresponding to a Moderately Demented severity state. In one embodiment herein, the representative structural brain MRI samples 606 represent structural brain MRI characteristics associated with the Moderately Demented severity state utilized for evaluation of advanced severity-state classification performed through the system 100. The representative structural brain MRI samples 606 represent structural brain MRI information associated with advanced cortical atrophy, enlarged ventricles, and pronounced hippocampal volume loss. The representative structural brain MRI samples 606 provide reference information associated with advanced severity-state analysis performed through the MRI acquisition and preprocessing module 110, deep neuroimaging feature encoder 112, and disease severity staging engine 114.

[0140] According to another example embodiment of the invention, FIG. 7 refers to a clinical-system integration architecture 700 for exchange of neuroimaging information, longitudinal patient information, and structured clinical decision-support records across interconnected healthcare computing environments. In one embodiment herein, the clinical-system integration architecture 700 represents integration of MRI acquisition resources, the Picture Archiving and Communication System (PACS) interface 132, neuroimaging processing resources associated with the system 100, the clinician decision interface 126, the Hospital Information System (HIS) 134, the hospital-edge processing node 136, and the cloud-synchronization service 138 for coordinated management of structural brain MRI information and longitudinal patient-state information.

[0141] In one embodiment herein, the clinical-system integration architecture 700 receives structural brain magnetic resonance imaging (MRI) information acquired through MRI acquisition resources and transfers the received MRI information through the PACS interface 132. The PACS interface 132 manages DICOM-formatted MRI information associated with patient-specific neuroimaging analysis cycles and facilitates availability of structural brain MRI information for processing through the system 100.

[0142] In one embodiment herein, the neuroimaging processing resources associated with the system 100 process received structural brain MRI information through the MRI acquisition and preprocessing module 110, the deep neuroimaging feature encoder 112, and the disease severity staging engine 114 for generating MRI-derived severity-state outputs. The clinical-system integration architecture 700 further manages prediction confidence information generated through the prediction uncertainty estimator 116, clinical consistency information generated through the clinical consistency validation engine 118, future severity-state information generated through the severity-state transition estimation engine 122, and spatial attribution information generated through the adaptive explainability manager 124.

[0143] In one embodiment herein, the hospital-edge processing node 136 provides a local processing environment associated with execution of neuroimaging analysis operations performed through the system 100. The hospital-edge processing node 136 supports processing of structural brain MRI information within a healthcare computing environment and facilitates generation of structured neuroimaging analysis information for clinician review through the clinician decision interface 126.

[0144] In one embodiment herein, the cloud-synchronization service 138 maintains synchronization of longitudinal patient-state information associated with the patient-specific predictive NeuroTwin 120. The cloud-synchronization service 138 supports storage and retrieval of longitudinal patient information comprises historical MRI analysis outcomes, severity-state history, prediction confidence information, clinical-consistency records, predicted future severity states, and severity-transition risk information.

[0145] In one embodiment herein, the clinician decision interface 126 receives consolidated neuroimaging analysis information generated through the system 100 and generates structured decision-support records. The structured decision-support records comprise MRI-derived severity-state information, prediction confidence information, spatial attribution information, clinical consistency information, and predicted future severity-state information for clinician review.

[0146] In one embodiment herein, the Hospital Information System (HIS) 134 stores structured severity-state information and clinical evidence information generated through the clinician decision interface 126. The clinical-system integration architecture 700 supports transfer of structural brain MRI information from MRI acquisition resources to the PACS interface 132, processing of received MRI information through the system 100, delivery of structured clinical decision-support information through the clinician decision interface 126, and storage of generated severity-state information and clinical evidence information within the Hospital Information System (HIS) 134.

[0147] The clinical-system integration architecture 700 establishes an integrated healthcare computing environment for managing MRI-based severity-state analysis, longitudinal patient-state maintenance, explainable neuroimaging generation, clinical consistency validation, severity-state transition estimation, and structured clinical decision-support generation through the system 100. The clinical-system integration architecture 700 supports coordinated exchange of neuroimaging information, longitudinal patient-state information, and structured decision-support records across connected healthcare computing environments.

[0148] According to another example embodiment of the invention, FIG. 8A refers to a graphical representation 800 of training and validation loss performance of the neuroimaging analysis system 100. In one embodiment herein, the graphical representation 800 represents convergence behaviour of the deep neuroimaging feature encoder 112 during training through the two-phase domain-adaptation protocol. The graphical representation 800 illustrates loss variation during a first training phase associated with frozen encoder layers and classification head optimization and a second training phase associated with selective adaptation of upper encoder layers for structural brain MRI data. During the first training phase, the classification head is trained using feature representations generated from frozen encoder layers for stabilizing classification performance. During the second training phase, upper encoder layers are selectively adapted to structural brain MRI data while normalization layers remain frozen for maintaining stable feature statistics during domain adaptation. The graphical representation 800 provides technical validation of adaptation of neuroimaging feature representations from a general image-processing domain to a structural brain MRI domain.

[0149] In one embodiment herein, Table 1 depicts overall experimental validation performance of the neuroimaging analysis system 100 evaluated using an Alzheimer's Multiclass MRI Dataset comprises 44,000 axial brain MRI images distributed across four severity states.

[0150] Table 1:

Clinical Metric	Value
Test Accuracy	78.82%
Macro F1-Score	0.789
Weighted F1-Score	0.782
Macro Precision	0.804
Macro Recall	0.791
Cohen's Kappa (κ)	0.715
AUC (macro, one-vs-rest)	>0.90 (all classes)
Total Validated Images	6,601 (test set)
Validation Framework	TensorFlow 2.x / Python 3.11

[0151] According to another example embodiment of the invention, FIG. 8B refers to a graphical representation 802 of training and validation accuracy performance of the neuroimaging analysis system 100. In one embodiment herein, the graphical representation 802 represents classification accuracy performance associated with the disease severity staging engine 114 using hierarchical neuroimaging feature representations generated through the deep neuroimaging feature encoder 112. The graphical representation 802 represents improvement in severity-state classification performance during training and validation cycles after domain adaptation of the deep neuroimaging feature encoder 112. The accuracy performance represents the capability of the disease severity staging engine 114 to generate MRI-derived severity-state outputs corresponding to Non-Demented, Very Mildly Demented, Mildly Demented, and Moderately Demented severity states.

[0152] In one embodiment herein, Table 2 depicts class-level severity-state classification performance generated by the disease severity staging engine 114.

[0153] Table 2:

Alzheimer's Severity Stage	Precision	Recall	F1-Score	Test Support
Non-Demented	0.6749	0.8709	0.7605	1,921
Very Mildly Demented	0.7312	0.5101	0.6010	1,680
Mildly Demented	0.8131	0.7833	0.7980	1,500
Moderately Demented	0.9953	0.9987	0.9970	1,500
Macro Average	0.8036	0.7908	0.7891	6,601
Weighted Average	0.7935	0.7882	0.7821	6,601

[0154] According to another example embodiment of the invention, FIG. 8C refers to a confusion matrix 804 representing classification performance across plurality of severity states. In one embodiment herein, the confusion matrix 804 represents comparison between severity-state outputs generated by the disease severity staging engine 114 and corresponding reference severity-state labels. The confusion matrix 804 provides class-level classification distribution information by representing correctly classified severity states and classification variations across Non-Demented, Very Mildly Demented, Mildly Demented, and Moderately Demented severity states.

[0155] The confusion matrix 804 further represents classification variations associated with adjacent severity states, particularly between Non-Demented and Very Mildly Demented severity states due to similarity in structural brain MRI characteristics. The classification variation information supports reliability assessment through the prediction uncertainty estimator 116 and clinical consistency validation through the clinical consistency validation engine 118. The confusion matrix 804 is generated using a 6,601-image test dataset.

[0156] According to another example embodiment of the invention, FIG. 8D refers to a graphical representation 806 of receiver operating characteristic (ROC) performance corresponding to severity-state classification. In one embodiment herein, the graphical representation 806 represents one-vs-rest ROC performance generated by the disease severity staging engine 114 for each predetermined severity state. The graphical representation 806 evaluates discrimination capability of the disease severity staging engine 114 by determining classification separation between a selected severity state and remaining severity states.

[0157] The graphical representation 806 demonstrates classification discrimination performance across all four severity states and supports evaluation of the capability of the disease severity staging engine 114 to generate reliable severity probability distributions from hierarchical neuroimaging feature representations.

[0158] In one embodiment herein, Table 3 depicts ROC-AUC performance associated with severity-state classification.

[0159] Table 3:

Severity State	ROC-AUC Performance
Non-Demented	>0.90
Very Mildly Demented	>0.90
Mildly Demented	>0.90
Moderately Demented	Approximately 1.00
Overall AUC (macro, one-vs-rest)	>0.90

[0160] According to another example embodiment of the invention, FIG. 8E refers to a graphical representation 808 of precision-recall performance corresponding to severity-state classification. In one embodiment herein, the graphical representation 808 represents precision and recall characteristics associated with MRI-derived severity-state outputs generated through the disease severity staging engine 114.

[0161] The graphical representation 808 evaluates classification reliability across different severity states and represents performance variation associated with severity-state boundaries. The precision-recall performance demonstrates high classification performance for Mildly Demented and Moderately Demented severity states, while Very Mildly Demented severity-state classification represents increased complexity due to subtle structural MRI variations associated with early-stage severity changes. The precision-recall information supports reliability evaluation through the prediction uncertainty estimator 116 and clinical consistency validation through the clinical consistency validation engine 118.

[0162] In one embodiment herein, Table 4 depicts precision-recall performance characteristics associated with severity-state classification.

[0163] Table 4:

Severity State	Observation
Non-Demented	Precision-recall characteristics evaluated using test dataset classification outcomes
Very Mildly Demented	Increased classification complexity due to subtle structural MRI variations
Mildly Demented	High precision-recall characteristics associated with severity-state classification
Moderately Demented	High precision-recall characteristics associated with severity-state classification

[0164] In one example embodiment, Table 5 depicts comparative benchmark information between the neuroimaging analysis system 100 and existing severity-classification approaches. In one embodiment herein, Table 5 compares technical characteristics associated with feature encoding, severity-state classification, clinical consistency validation, progression forecasting, adaptive explainability, and prediction uncertainty estimation.

[0165] Table 5:

Clinical Staging System	Feature Encoder	Classes	Accuracy	Clinical Validation Engine	Progression Forecast	Adaptive Explainability	Uncertainty Estimation
Farooq et al. (2017)	AlexNet	4	74.2%	None	None	None	None
Ebrahimi-Ghahnavieh et al. (2020)	VGG19	4	72.8%	None	None	None	None
Liu et al. (2022)	ResNet-50	4	76.1%	None	None	Fixed only	None
NeuroTwin-X (System 100)	Deep Neuroimaging Encoder (two-	4	78.82%	Clinical Consistency Score Engine	Severity-State Transition	Adaptive Explainability Manager	Prediction Uncertainty

	phase)				Estimatio n Engine		Estimato

[0166] According to another example embodiment of the invention, FIGs. 9A-9B refers to a flowchart 900 of a method for training and operating the system 100 for processing structural brain magnetic resonance imaging (MRI) data, generating severity-state information, determining prediction confidence information, performing clinical consistency validation, generating future severity-state information, updating longitudinal patient-state information, and retrieving updated information during a subsequent MRI analysis cycle. In one embodiment herein, the flowchart 900 represents a method workflow executed through the system 100 for generating MRI-derived severity-state outputs, confidence-responsive explainability information, clinical consistency information, severity-transition information, and structured decision-support records.

[0167] At step 902, the method trains the deep neuroimaging feature encoder 112 through a two-phase domain-adaptation protocol. In one embodiment herein, the deep neuroimaging feature encoder 112 is obtained from a deep convolutional, compound-scaled convolutional, attention-augmented convolutional, or attention-based feature encoder pretrained on a natural-image dataset. During a first training phase, encoder layers of the deep neuroimaging feature encoder 112 remain frozen and a classification head coupled to the deep neuroimaging feature encoder 112 is trained using feature representations generated by the frozen encoder layers on a labelled structural brain MRI dataset. During a second training phase, upper encoder layers of the deep neuroimaging feature encoder 112 are selectively unfrozen and adapted to the structural brain MRI dataset while normalization layers remain frozen, thereby producing a domain-adapted deep neuroimaging feature encoder 112.

[0168] At step 904, the computing device 102 receives previously acquired structural brain magnetic resonance imaging (MRI) data through the MRI acquisition and preprocessing module 110. In one embodiment herein, the MRI acquisition and preprocessing module 110 receives structural brain MRI data from an imaging source through clinical imaging infrastructure for subsequent neuroimaging processing. At step 906, the MRI acquisition and preprocessing module 110 performs image conversion, intensity normalization, and spatial resampling on the structural brain MRI data to generate standardized neuroimaging input data. In one embodiment herein, the MRI acquisition and preprocessing module 110 processes Digital Imaging and Communications in Medicine (DICOM)-formatted structural brain MRI data received through the Picture Archiving and Communication System (PACS) interface 132 and generates standardized neuroimaging input data for feature extraction.

[0169] At step 908, the domain-adapted deep neuroimaging feature encoder 112 processes the standardized neuroimaging input data and generates a hierarchical neuroimaging feature representation. In one embodiment herein, the hierarchical neuroimaging feature representation represents extracted structural characteristics associated with the processed structural brain MRI data for subsequent severity-state analysis. At step 910, the disease severity staging engine 114 maps the hierarchical neuroimaging feature representation to a severity probability distribution across plurality of predetermined severity states and generates an MRI-derived severity-state output. In one embodiment herein, the predetermined severity states correspond to Non-Demented, Very Mildly Demented, Mildly Demented, and Moderately Demented severity states.

[0170] At step 912, the prediction uncertainty estimator 116 determines prediction confidence information from the severity probability distribution. In one embodiment herein, the prediction uncertainty estimator 116 evaluates uncertainty associated with the severity probability distribution through entropy of the severity probability distribution, variance obtained through Monte Carlo sampling, disagreement among inference model outputs, or combinations thereof. At step 914, the prediction uncertainty estimator 116 compares the prediction confidence information with a confidence threshold and generates an escalation flag responsive to the prediction confidence information failing to satisfy the confidence threshold. In one embodiment herein, the escalation flag represents reliability-related information associated with the MRI-derived severity-state output and supports subsequent clinical review operations.

[0171] At step 916, the adaptive explainability manager 124 dynamically selects a spatial attribution technique based on the prediction confidence information and generates spatial attribution information corresponding to the MRI-derived severity-state output using the selected spatial attribution technique. In one embodiment herein, the adaptive explainability manager 124 selects a gradient-weighted spatial attribution technique responsive to a high-confidence condition, a score-based activation attribution technique responsive to an intermediate-confidence condition, and a layer-wise composite attribution technique responsive to a low-confidence condition. At step 918, the clinical consistency validation engine 118 receives the MRI-derived severity-state output and one or more structured patient-specific clinical parameters. In one embodiment herein, the structured patient-specific clinical parameters comprise patient age, Mini-Mental State Examination (MMSE) score, Clinical Dementia Rating (CDR) score, prior MRI-derived severity state, clinical-history indicator, and apolipoprotein E (APOE) genotype indicator.

[0172] At step 920, the clinical consistency validation engine 118 determines concordance between the MRI-derived severity-state output and the structured patient-specific clinical parameters. In one embodiment herein, the clinical consistency validation engine 118 evaluates agreement between imaging-derived severity-state information and available structured clinical parameters through concordance evaluation. At step 922, the clinical consistency validation engine 118 generates a clinical consistency score representing the determined concordance. In one embodiment herein, the clinical consistency validation engine 118 generates the clinical consistency score through weighted aggregation of concordance values associated with available structured patient-specific clinical parameters and normalizes the generated score to a predefined scoring range.

[0173] At step 924, the clinical consistency validation engine 118 compares the clinical consistency score with a consistency threshold and generates a discordance alert responsive to the clinical consistency score failing to satisfy the consistency threshold. In one embodiment herein, the discordance alert identifies inconsistency between MRI-derived severity-state information and structured patient-specific clinical parameters for further clinical review. At step 926, the system 100 retrieves longitudinal information associated with one or more preceding MRI analysis cycles from the patient-specific predictive NeuroTwin 120 stored in the longitudinal patient data store 108. In one embodiment herein, the retrieved longitudinal information comprises historical MRI analysis outcomes, longitudinal severity-state history, prediction confidence information, clinical-consistency records, treatment-response records, predicted future severity states, and severity-transition risk information.

[0174] At step 928, the severity-state transition estimation engine 122 processes the MRI-derived severity-state output together with the longitudinal information retrieved from the patient-specific predictive NeuroTwin 120 to generate a future severity-state probability distribution and severity-transition risk information. In one embodiment herein, the severity-state transition estimation engine 122 evaluates current severity-state information, longitudinal severity-state history, assessment intervals, patient age, and structured clinical parameters. The severity-state transition estimation engine 122 further generates a severity-transition risk score representing probability of transition from a current severity state to a next or higher severity state within a selected future time horizon and generates longitudinal risk information associated with patient-specific severity-state progression.

[0175] At step 930, the system 100 stores a predicted future severity state in the patient-specific predictive NeuroTwin 120. In one embodiment herein, the patient-specific predictive NeuroTwin 120 stores future severity-state probability information, severity-transition risk information, and generated predicted future-state information as a longitudinal patient-state entry. At step 932, the clinician decision interface 126 generates a structured neuroimaging analysis record consolidating the MRI-derived severity-state output, the prediction confidence information, the spatial attribution information, the clinical consistency score, and the predicted future severity state. In one embodiment herein, the structured neuroimaging analysis record is stored in the longitudinal patient data store 108 and presented through the clinician decision interface 126 for clinician review.

[0176] At step 934, the at least one processor 104 updates the patient-specific predictive NeuroTwin 120 with the MRI-derived severity-state output, the prediction confidence information, the clinical consistency score, and the predicted future severity state. In one embodiment herein, the update operation maintains longitudinal patient-state information associated with successive MRI analysis cycles. At step 936, the system 100 retrieves information from the updated patient-specific predictive NeuroTwin 120 during a subsequent MRI analysis cycle for processing structural brain MRI data associated with the subsequent MRI analysis cycle. In one embodiment herein, the retrieved longitudinal patient-state information supports generation of updated severity-state information, prediction confidence information, clinical consistency information, and future severity-state information during subsequent analysis cycles.

[0177] In one embodiment herein, the method further comprises generating a structured escalation-protocol entry responsive to the escalation flag or the discordance alert. The structured escalation-protocol entry identifies at least one of an additional-imaging action or a specialist-referral action for inclusion within the structured decision-support record generated through the clinician decision interface 126. In one embodiment herein, the method further comprises exchanging neuroimaging information and longitudinal patient information through the communication interface 128 and the network 130 with one or more of the PACS interface 132, the Hospital Information System (HIS) 134, the hospital-edge processing node 136, and the cloud-synchronization service 138. The communication operation supports transfer and synchronization of structured neuroimaging analysis information and longitudinal patient-state information across interconnected healthcare computing environments.

[0178] The flowchart 900 establishes a method for training and operating the system 100 by integrating domain-adapted neuroimaging feature extraction, MRI-based severity-state classification, prediction confidence evaluation, adaptive explainability generation, clinical consistency validation, longitudinal NeuroTwin management, severity-state transition estimation, and structured clinical decision-support generation. The method supports repeated MRI analysis cycles through continuous updating and retrieval of patient-specific longitudinal neuroimaging information.

[0179] Numerous advantages of the present disclosure may be apparent from the discussion above. In accordance with the present disclosure, a system 100 for MRI-based alzheimer’s severity grading, clinical consistency validation, progression forecasting is disclosed. The proposed invention provides a system 100 for processing structural brain magnetic resonance imaging (MRI) data to generate multi-stage Alzheimer’s severity probability information, cross-source clinical consistency information, patient-specific severity-state transition estimates, and confidence-responsive spatial attribution information. The system 100 standardizes structural brain MRI data through image conversion, intensity normalization, and spatial resampling for subsequent neuroimaging analysis. The system 100 extracts hierarchical neuroimaging feature representations through staged domain adaptation involving initial feature extraction and subsequent selective adaptation of a deep neuroimaging feature encoder to structural brain MRI data. The system 100 maps neuroimaging feature representations to multi-stage Alzheimer’s severity probability information for characterization of MRI-derived severity states. The system 100 determines prediction confidence information from output-distribution characteristics, sampling variance, model disagreement, or one or more corresponding uncertainty indicators and generates an escalation indication upon satisfaction of a configurable confidence condition. The system 100 cross-evaluates MRI-derived severity information against structured patient-specific clinical parameters and generates quantified clinical consistency information together with a discordance indication based on a configurable consistency condition.

[0180] The system 100 creates and updates a Patient-Specific Predictive NeuroTwin storing an imaging timeline, severity-state history, predicted future states, clinical consistency history, treatment-response records, and severity-transition risk information across successive analysis cycles. The system 100 processes longitudinal severity-state history, assessment intervals, patient age, and structured clinical parameters to estimate future severity-state probabilities and corresponding transition-risk information. The system 100 selects a spatial attribution technique in response to prediction confidence information and generates MRI-based attribution evidence together with annotated anatomical-region information for interpretation of an MRI-derived severity output. The system 100 supports reduction of inter-rater variability and improvement of longitudinal consistency through coordinated processing of current MRI-derived severity information, prior assessment information, prediction confidence information, and structured patient-specific clinical parameters. The system 100 supports interoperable data exchange among MRI acquisition equipment, Picture Archiving and Communication Systems (PACS), neuroimaging analysis resources, clinician review interfaces, Hospital Information Systems (HIS), hospital-edge processing resources, and cloud-synchronization resources for longitudinal analysis and low-latency processing. The system 100 supports modular replacement of a neuroimaging feature encoder and integration of modality-specific feature streams derived from electroencephalography, speech, or eye-tracking data while maintaining interoperability with downstream severity-state, consistency, and longitudinal analysis operations. The system 100 supports reproducible and auditable neuroimaging processing through model-version tracking, class-level performance documentation, structured processing records, and traceable longitudinal analysis information.

[0181] It will readily be apparent that numerous modifications and alterations can be made to the processes described in the foregoing examples without departing from the principles underlying the invention, and all such modifications and alterations are intended to be embraced by this application.

5. CLAIMS:

I/We Claim:

1. A system (100) for processing structural brain magnetic resonance imaging (MRI) data and maintaining patient-specific longitudinal neuroimaging state information, comprising:

a computing device (102) having at least one processor (104) and a memory (106) operatively coupled to the at least one processor (104), the memory (106) storing executable instructions;

a longitudinal patient data store (108) accessible to the at least one processor (104);

a magnetic resonance imaging (MRI) acquisition and preprocessing module (110) configured to receive previously acquired structural brain MRI data from an imaging source and transform the structural brain MRI data into standardized neuroimaging input data through image conversion, intensity normalization, and spatial resampling;

a deep neuroimaging feature encoder (112) configured to process the standardized neuroimaging input data and generate a hierarchical neuroimaging feature representation, wherein the deep neuroimaging feature encoder (112) is domain-adapted through a two-phase domain-adaptation protocol that comprises:

> a first training phase in which encoder layers of the deep neuroimaging feature encoder (112) remain frozen while a classification head coupled to the encoder layers is trained using feature representations generated by the frozen encoder layers; and
>
> a second training phase in which upper encoder layers of the deep neuroimaging feature encoder (112) are selectively unfrozen and adapted to structural brain MRI data while normalization layers remain frozen;

a disease severity staging engine (114) configured to map the hierarchical neuroimaging feature representation to a severity probability distribution across plurality of predetermined severity states and generate an MRI-derived severity-state output;

a prediction uncertainty estimator (116) configured to determine prediction confidence information from the severity probability distribution, compare the prediction confidence information with a confidence threshold stored in the memory (106), and generate an escalation flag responsive to the prediction confidence information failing to satisfy the confidence threshold;

a clinical consistency validation engine (118) configured to receive the MRI-derived severity-state output and one or more structured patient-specific clinical parameters, determine concordance between the MRI-derived severity-state output and the one or more structured patient-specific clinical parameters, generate a clinical consistency score representing the determined concordance, and generate a discordance alert responsive to the clinical consistency score failing to satisfy a consistency threshold stored in the memory (106);

a patient-specific predictive NeuroTwin (120), maintained in the longitudinal patient data store (108), configured to maintain historical MRI analysis outcomes, a longitudinal severity-state history, prediction confidence information, clinical-consistency records, treatment-response records, predicted future severity states, and severity-transition risk information;

a severity-state transition estimation engine (122) configured to process a current MRI-derived severity-state output together with longitudinal information retrieved from the patient-specific predictive NeuroTwin (120), generate a future severity-state probability distribution and severity-transition risk information, and store a predicted future severity state in the patient-specific predictive NeuroTwin (120);

an adaptive explainability manager (124) configured to receive the prediction confidence information from the prediction uncertainty estimator (116), dynamically select a spatial attribution technique from plurality of distinct spatial attribution techniques according to the prediction confidence information, and generate spatial attribution information corresponding to the MRI-derived severity-state output using the selected spatial attribution technique; and a clinician decision interface (126) configured to consolidate the MRI-derived severity-state output, the prediction confidence information, the spatial attribution information, the clinical consistency score, and the predicted future severity state into a structured decision-support record for clinician review,

wherein the at least one processor (104) is configured to update the patient-specific predictive NeuroTwin (120), after an MRI analysis cycle, using the MRI-derived severity-state output, the prediction confidence information, the clinical consistency score, and the predicted future severity state, and retrieve information from the updated patient-specific predictive NeuroTwin (120) during a subsequent MRI analysis cycle, and

wherein the clinical consistency validation engine (118) and the severity-state transition estimation engine (122) process longitudinal information accumulated across successive MRI analysis cycles.

2. The system (100) as claimed in claim 1, wherein the prediction uncertainty estimator (116) is configured to derive the prediction confidence information from at least one of entropy of the severity probability distribution, variance obtained through Monte Carlo sampling, and disagreement among outputs generated by plurality of inference models.

3. The system (100) as claimed in claim 1, wherein the MRI acquisition and preprocessing module (110) is configured to receive Digital Imaging and Communications in Medicine (DICOM)-formatted structural brain MRI data through a Picture Archiving and Communication System (PACS) interface (132), perform DICOM-to-tensor conversion, and generate the standardized neuroimaging input data, and

wherein the plurality of predetermined severity states comprises a Non-Demented severity state, a Very Mildly Demented severity state, a Mildly Demented severity state, and a Moderately Demented severity state.

4. The system (100) as claimed in claim 1, wherein the one or more structured patient-specific clinical parameters comprise at least one of patient age, a Mini-Mental State Examination (MMSE) score, a Clinical Dementia Rating (CDR) score, a prior MRI-derived severity state, a clinical-history indicator, and an apolipoprotein E (APOE) genotype indicator, and

wherein the clinical consistency validation engine (118) is configured to:

generate, for each available structured patient-specific clinical parameter, a concordance value representing agreement between the respective structured patient-specific clinical parameter and an expected clinical profile corresponding to the MRI-derived severity-state output;

apply a respective reliability weight to each generated concordance value;

generate the clinical consistency score by weighted aggregation of the concordance values followed by normalization to a scoring range from 0 to 100; and

generate the discordance alert responsive to the clinical consistency score being below the consistency threshold,

wherein each concordance value comprises a binary concordance value representing a concordant state or a discordant state, and

wherein the consistency threshold has an exemplary default value of 60 on the scoring range from 0 to 100.

5. The system (100) as claimed in claim 1, wherein the adaptive explainability manager (124) is configured to:

select a gradient-weighted spatial attribution technique responsive to the prediction confidence information being at least 85 percent;

select a score-based activation attribution technique responsive to the prediction confidence information being from 70 percent to less than 85 percent; and

select a layer-wise composite attribution technique responsive to the prediction confidence information being below 70 percent,

wherein the spatial attribution information comprises a structural brain MRI image, a spatial attribution map corresponding to the MRI-derived severity-state output, and an anatomical-region overlay associated with the spatial attribution information.

6. The system (100) as claimed in claim 1, wherein the severity-state transition estimation engine (122) is configured to;

retrieve, from the patient-specific predictive NeuroTwin (120), a current severity state, a chronological severity-state history, time intervals between successive MRI analysis cycles, patient age, and one or more structured patient-specific clinical parameters;

apply a probabilistic stage-transition model that comprises a learned transition-probability matrix conditioned on the retrieved current severity state, the chronological severity-state history, the time intervals between successive MRI analysis cycles, the patient age, and the one or more structured patient-specific clinical parameters;

generate probabilities corresponding to respective future severity states at a selected future time horizon;

generate a severity-transition risk score representing a probability of transition from the current severity state to a next or higher severity state within the selected future time horizon; and

store the probabilities corresponding to the respective future severity states and the severity-transition risk score as a predicted future-state entry in the patient-specific predictive NeuroTwin (120).

7. The system (100) as claimed in claim 1, wherein the system (100) comprises a communication interface (128) configured to exchange neuroimaging information, longitudinal patient information, and structured decision-support information through a network (130) with one or more of a Picture Archiving and Communication System (PACS) interface (132), a Hospital Information System (HIS) (134), a hospital-edge processing node (136), and a cloud-synchronization service (138),

wherein the hospital-edge processing node (136) provides a local processing environment for processing structural brain MRI information, and

wherein the cloud-synchronization service (138) synchronizes longitudinal patient-state information associated with the patient-specific predictive NeuroTwin (120) for storage and subsequent retrieval.

8. The system (100) as claimed in claim 1, wherein the deep neuroimaging feature encoder (112) communicates the hierarchical neuroimaging feature representation to the disease severity staging engine (114) through a structured feature interface configured to receive feature representations generated by a deep convolutional neuroimaging encoder or an attention-based neuroimaging encoder, and

wherein the deep neuroimaging feature encoder (112) is replaceable while maintaining interoperability with downstream modules through structured exchange of generated information.

9. The system (100) as claimed in claim 1, wherein the standardized neuroimaging input data comprises an input representation having dimensions of 224 × 224 × 3, and wherein the deep neuroimaging feature encoder (112) comprises an EfficientNetB0 backbone pretrained using an ImageNet dataset,

wherein feature information generated by the deep neuroimaging feature encoder (112) undergoes global average pooling before being supplied to a classification head that comprises a dense layer having 256 units and a rectified-linear-unit activation, a batch-normalization operation, a dropout operation having a dropout value of 0.4, and a SoftMax output corresponding to the plurality of predetermined severity states.

10. The system (100) as claimed in claim 6, wherein the selected future time horizon is six months, and wherein the probabilities corresponding to the respective future severity states comprise probabilities corresponding to the Non-Demented severity state, the Very Mildly Demented severity state, the Mildly Demented severity state, and the Moderately Demented severity state.

6. DATE AND SIGNATURE:

Dated this 27th day of August, 2026

Patent Agent Name: Hima Bindu Atti

INPA - 3925

7. ABSTRACT:

Title: System for MRI-Based Alzheimer's Severity Grading, Clinical Consistency Validation, Progression Forecasting, and Method Thereof

The present disclosure proposes a system (100) and method for processing structural brain MRI data for multi-stage Alzheimer's severity grading, clinical consistency validation, patient-specific severity-state transition estimation, and adaptive explainable neuroimaging. The system (100) comprises a computing device (102), a longitudinal patient data store 108, an MRI acquisition and preprocessing module (110), a deep neuroimaging feature encoder (112), a disease severity staging engine (114), a prediction uncertainty estimator (116), a clinical consistency validation engine (118), a patient-specific predictive NeuroTwin (120), a severity-state transition estimation engine (122), an adaptive explainability manager (124), a clinician decision interface (126), a communication interface (128), a network (130). The system (100) supports reproducible and auditable neuroimaging processing through model-version tracking, class-level performance documentation, structured processing records, and traceable longitudinal analysis information.

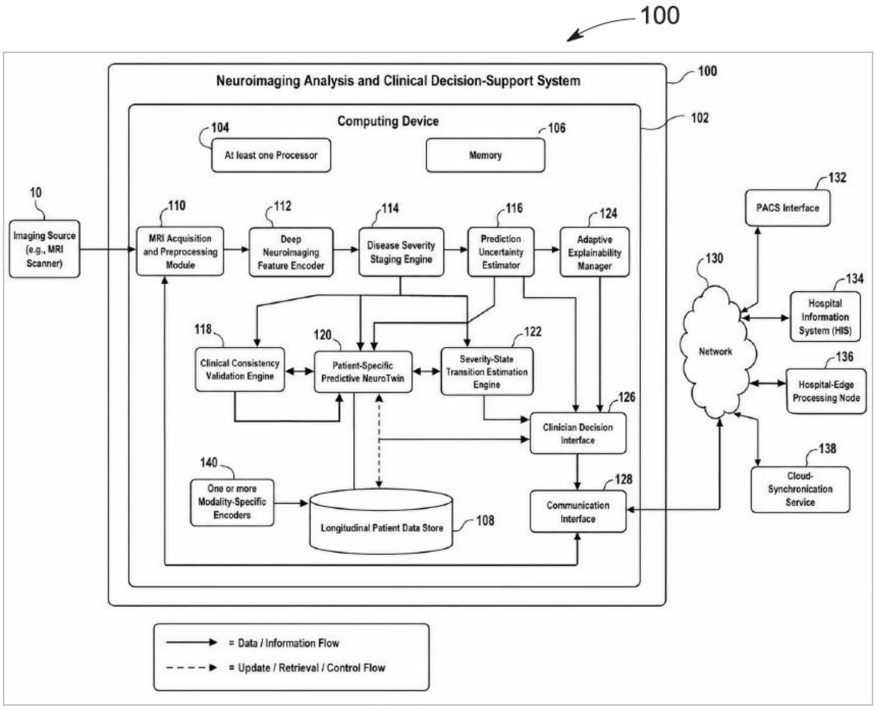

FIG. 1

Digitally Signed By:
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Date: 27-08-2026 13:14:10